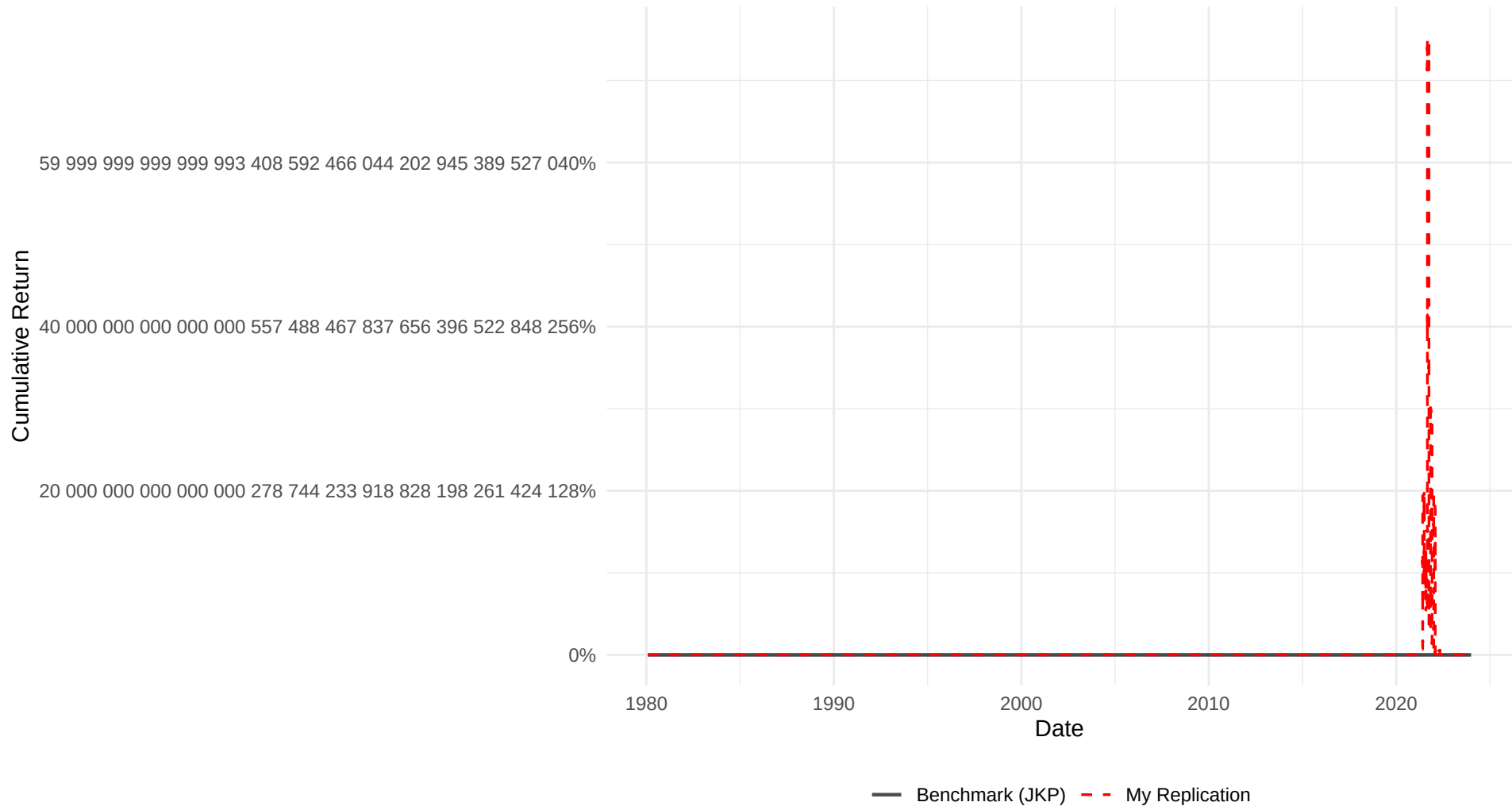


Monthly Corr: -0.4263 | Drift Check (Red should overlap Black)



Factor: betabab_1260d (Cur

Monthly Corr: -0.3706 | Drift Check (P

Cumulative Return

249 999 999 999 999 989 854 181 114 087 590 682 872 999 022 412 112 859 917 434 752 451 675 980 737 738 606 379 008%
200 000 000 000 000 002 926 139 046 134 974 606 194 008 597 572 931 011 855 722 157 433 959 272 850 229 643 182 080%
150 000 000 000 000 000 015 998 096 978 182 358 529 515 018 172 733 749 163 794 009 562 416 242 564 962 720 679 985 152%
100 000 000 000 000 000 001 463 069 523 067 487 303 097 004 298 786 465 505 927 861 078 716 979 636 425 114 821 591 040%
50 000 000 000 000 000 000 731 534 761 533 743 651 548 502 149 393 232 752 963 930 539 358 489 818 212 557 410 795 520%
0%

1980 1990 2000 2010 2020

Date

Benchmark (JKP) My Replication

Factor: beta_60

Monthly Corr: -0.358

Cumulative Return

74 999 999 999 999 982 737 362 357 087 567 679 777 817 949 406 159 997 320 494 985 687 783 338 899 059 965 436 510 315 444 445 380 608%

49 999 999 999 999 998 852 475 663 262 266 831 422 342 135 996 207 500 306 499 798 736 599 672 609 039 495 565 163 064 724 075 577 344%

24 999 999 999 999 999 426 237 831 631 133 415 711 171 067 998 103 750 153 249 899 368 299 836 304 519 747 782 581 532 362 037 788 672%

0%

19801990200020102020

Date

— Benchmark (JKP) - - My R

Factor: qmj_safety (Cu
Monthly Corr: -0.3526 | Drift C

Cumulative Return

149 999 999 999 999 977 558 572 236 863 829 055 100 797 746 570 813 933 451 269 901 990 276 512 069 056 164 662 916 153 344%

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

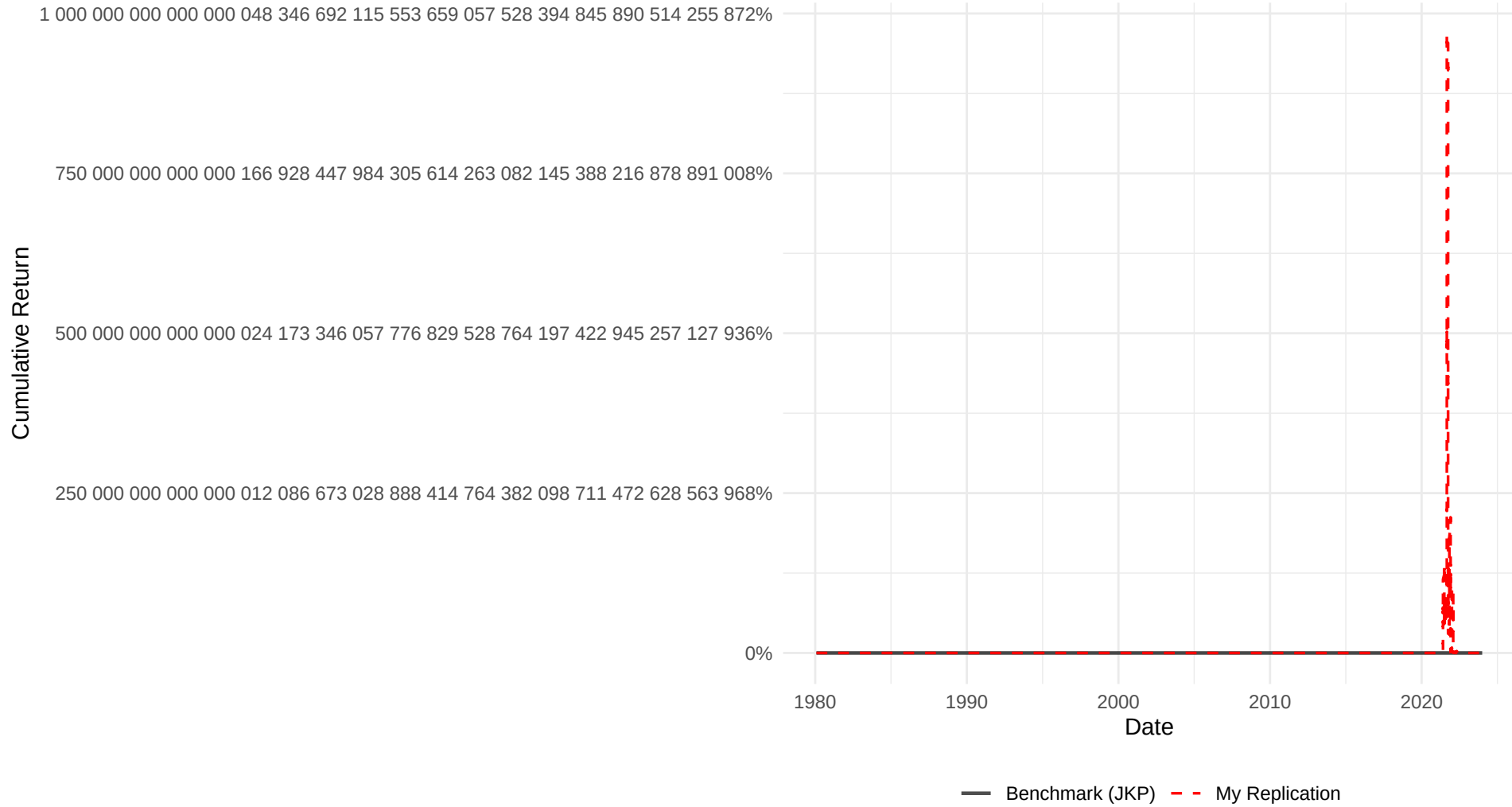
0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replicat

Factor: ivol_ff3_21d (Cumulative Wealth)

Monthly Corr: -0.3485 | Drift Check (Red should overlap Black)



Factor: betadown_252d (Cumulative)

Monthly Corr: -0.3319 | Drift Check (Red s

Cumulative Return

39 999 999 999 999 998 536 271 862 523 546 296 844 108 572 901 094 271 174 721 455 373 708 346 006 171 549 696%

29 999 999 999 999 998 902 203 896 892 659 722 633 081 429 675 820 703 381 041 091 530 281 259 504 628 662 272%

19 999 999 999 999 999 268 135 931 261 773 148 422 054 286 450 547 135 587 360 727 686 854 173 003 085 774 848%

9 999 999 999 999 999 634 067 965 630 886 574 211 027 143 225 273 567 793 680 363 843 427 086 501 542 887 424%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: earnings_variability

Monthly Corr: -0.3224 | Drift Check

Cumulative Return

15 000 000 000 000 000 002 041 521 464 012 831 935 347 326 192 459 104 545 336 271 930 460 528 996 026 673 618 321 670 144%

9 999 999 999 999 999 999 594 167 244 563 503 627 314 919 960 896 484 514 396 697 390 098 067 039 229 509 544 255 160 320%

4 999 999 999 999 999 999 797 083 622 281 751 813 657 459 980 448 242 257 198 348 695 049 033 519 614 754 772 127 580 160%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: beta_dimson

Monthly Corr: -0.3119 | Drift

Cumulative Return

2 499 999 999 999 999 587 378 840 868 625 085 777 156 300 773 284 262 939 764 503 477 957 753 854 665 074 741 744 226 009 088%

1 999 999 999 999 999 855 170 415 474 605 981 299 438 616 632 528 062 917 043 578 247 391 105 546 864 194 206 056 388 231 168%

1 499 999 999 999 999 891 377 811 605 954 485 974 578 962 474 396 047 187 782 683 685 543 329 160 148 145 654 542 291 173 376%

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: prc_highprc_252d (Cu

Monthly Corr: -0.3107 | Drift Check (Re

Cumulative Return

9 999 999 999 999 998 420 870 365 609 107 783 451 011 464 309 390 187 480 008 864 829 101 324 851 880 426 209 280%

4 999 999 999 999 999 210 435 182 804 553 891 725 505 732 154 695 093 740 004 432 414 550 662 425 940 213 104 640%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: rvol_21d (Cumula

Monthly Corr: -0.3101 | Drift Chec

Cumulative Return

399 999 999 999 999 997 901 466 300 767 219 729 263 178 441 802 728 702 487 766 778 927 633 234 153 231 380 547 371 008%

299 999 999 999 999 956 021 770 170 894 190 886 948 243 813 522 002 147 006 211 559 180 869 930 696 374 704 387 653 632%

199 999 999 999 999 998 950 733 150 383 609 864 631 589 220 901 364 351 243 883 389 463 816 617 076 615 690 273 685 504%

99 999 999 999 999 999 475 366 575 191 804 932 315 794 610 450 682 175 621 941 694 731 908 308 538 307 845 136 842 752%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: zero_trades_25

Monthly Corr: -0.3029 | Drift Coef: 0.0000

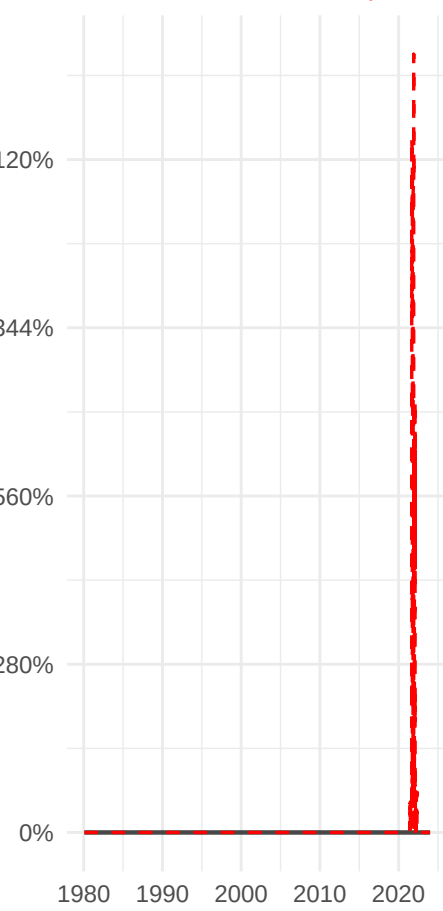
Cumulative Return

199 999 999 999 999 979 727 437 085 594 788 358 765 312 412 818 410 899 040 858 591 457 082 352 713 540 220 209 982 341 120%

149 999 999 999 999 977 558 572 236 863 829 055 100 797 746 570 813 933 451 269 901 990 276 512 069 056 164 662 916 153 344%

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%



— Benchmark (JKP) - - My Replication

Factor: qmj (Cumulat

Monthly Corr: -0.2995 | Drift

Cumulative Return

1 999 999 999 999 999 855 170 415 474 605 981 299 438 616 632 528 062 917 043 578 247 391 105 546 864 194 206 056 388 231 168%

1 499 999 999 999 999 891 377 811 605 954 485 974 578 962 474 396 047 187 782 683 685 543 329 160 148 145 654 542 291 173 376%

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: zero_trades_21d (Cur

Monthly Corr: -0.2986 | Drift Check (Re

Cumulative Return

2 999 999 999 999 999 957 620 256 357 142 571 749 975 680 685 131 147 022 196 970 209 379 557 153 221 891 850 240%

1 999 999 999 999 999 684 174 073 121 821 556 690 202 292 861 878 037 496 001 772 965 820 264 970 376 085 241 856%

999 999 999 999 999 842 087 036 560 910 778 345 101 146 430 939 018 748 000 886 482 910 132 485 188 042 620 928%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: turnover_126d (

Monthly Corr: -0.2972 | Drift CH

Cumulative Return

30 000 000 000 000 000 577 618 351 505 349 360 801 390 143 444 258 755 270 815 809 519 829 979 140 080 533 278 782 652 416%

19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%

9 999 999 999 999 998 986 371 854 279 739 417 938 265 620 640 920 544 952 042 929 572 854 117 635 677 011 010 499 117 056%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: zero_trades_126

Monthly Corr: -0.2955 | Drift Change: 0.0000

Cumulative Return

7 500 000 000 000 000 144 404 587 876 337 340 200 347 535 861 064 688 817 703 952 379 957 494 785 020 133 319 695 663 104%

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

2 499 999 999 999 999 746 592 963 569 934 854 484 566 405 160 230 136 238 010 732 393 213 529 408 919 252 752 624 779 264%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: rmax5_21d (Cumulative)

Monthly Corr: -0.2922 | Drift CH

Cumulative Return

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

74 999 999 999 999 988 779 286 118 431 914 527 550 398 873 285 406 966 725 634 950 995 138 256 034 528 082 331 458 076 672%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

24 999 999 999 999 997 465 929 635 699 348 544 845 664 051 602 301 362 380 107 323 932 135 294 089 192 527 526 247 792 640%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: ivol_capm_21d (Cur

Monthly Corr: -0.29 | Drift Check (Re

Cumulative Return

400 000 000 000 000 000 005 852 278 092 269 949 212 388 017 195 145 862 023 711 444 314 867 918 545 700 459 286 364 160%

200 000 000 000 000 000 002 926 139 046 134 974 606 194 008 597 572 931 011 855 722 157 433 959 272 850 229 643 182 080%

0%

1980 1990 2000 2010 2020

Date

Benchmark (JKP) My Replication

Factor: rmax1_21d (Cumulative)

Monthly Corr: -0.289 | Drift Check: 0.000

Cumulative Return

24 999 999 999 999 997 465 929 635 699 348 544 845 664 051 602 301 362 380 107 323 932 135 294 089 192 527 526 247 792 640%

19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%

15 000 000 000 000 000 288 809 175 752 674 680 400 695 071 722 129 377 635 407 904 759 914 989 570 040 266 639 391 326 208%

9 999 999 999 999 998 986 371 854 279 739 417 938 265 620 640 920 544 952 042 929 572 854 117 635 677 011 010 499 117 056%

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replication

Factor: ivol_capm_252d (Cumulative)

Monthly Corr: -0.2845 | Drift Check (R-squared)



Factor: niq_be (Cumulative Wealth)

Monthly Corr: -0.2566 | Drift Check (Red should overlap Bla

Cumulative Return

39 999 999 999 999 999 309 471 031 356 742 393 268 959 131 502 323 729 114 308 588 601 344%

30 000 000 000 000 000 978 680 950 144 401 383 192 292 617 328 216 608 963 406 365 458 432%

19 999 999 999 999 999 654 735 515 678 371 196 634 479 565 751 161 864 557 154 294 300 672%

9 999 999 999 999 999 827 367 757 839 185 598 317 239 782 875 580 932 278 577 147 150 336%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication

Factor: f_score (Cumulative Wealth)

Monthly Corr: -0.2472 | Drift Check (Red shows)

15 000 000 000 000 001 978 596 948 491 702 342 066 573 379 246 000 727 344 002 835 378 319 299 855 777 792%

10 000 000 000 000 001 319 064 632 327 801 561 377 715 586 164 000 484 896 001 890 252 212 866 570 518 528%

5 000 000 000 000 000 659 532 316 163 900 780 688 857 793 082 000 242 448 000 945 126 106 433 285 259 264%

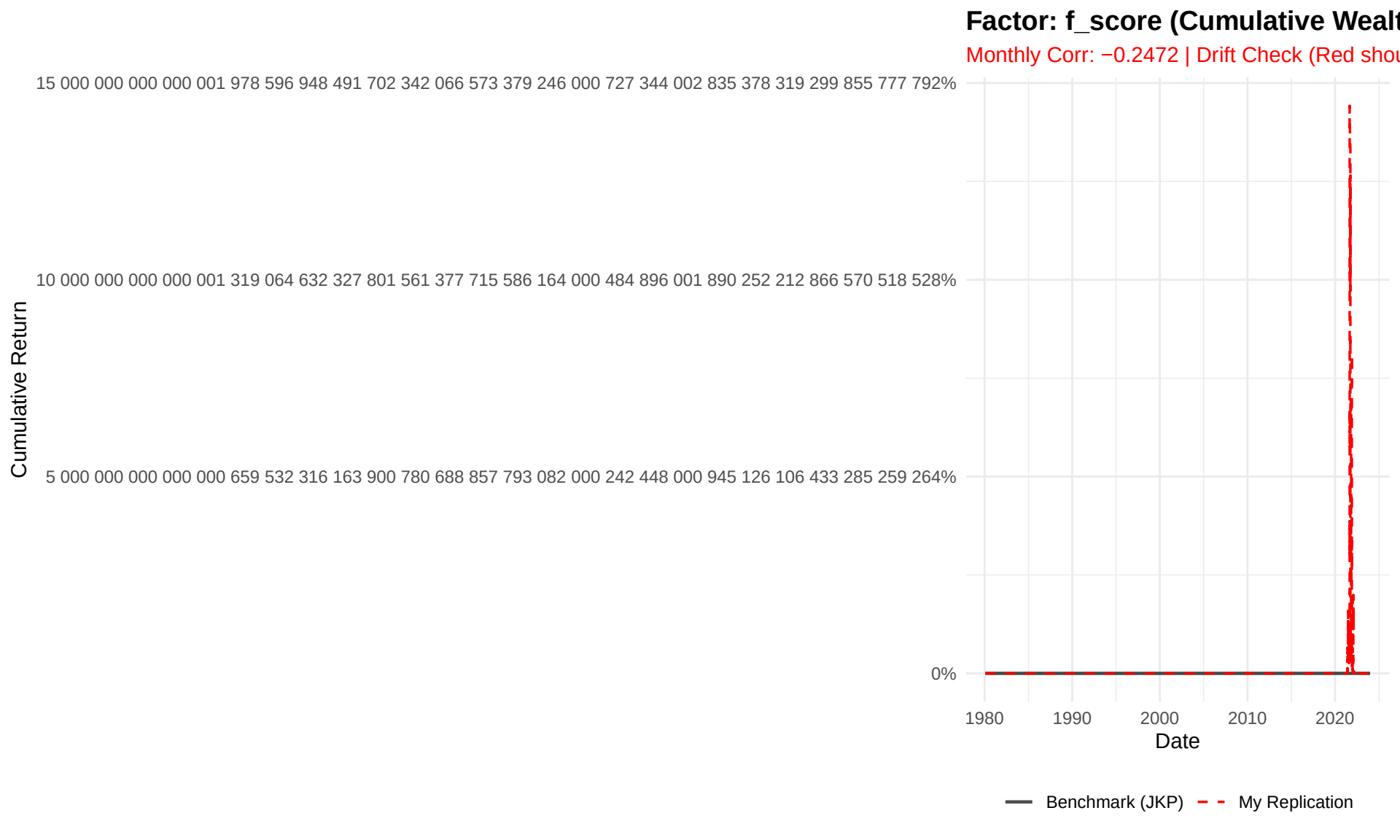
0%

Cumulative Return

1980 1990 2000 2010 2020

Date

Benchmark (JKP) My Replication



Factor: ebit_sale (Cumula

Monthly Corr: -0.2391 | Drift Chec

Cumulative Return

599 999 999 999 999 912 043 540 341 788 381 773 896 487 627 044 004 294 012 423 118 361 739 861 392 749 408 775 307 264%

399 999 999 999 999 997 901 466 300 767 219 729 263 178 441 802 728 702 487 766 778 927 633 234 153 231 380 547 371 008%

199 999 999 999 999 998 950 733 150 383 609 864 631 589 220 901 364 351 243 883 389 463 816 617 076 615 690 273 685 504%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: eqnpo_me (Cumulative Wealth

Monthly Corr: -0.2365 | Drift Check (Red should o

Cumulative Return

1 999 999 999 999 999 853 079 562 352 962 397 847 017 606 430 398 935 774 525 292 839 560 724 611 072%

999 999 999 999 999 999 926 539 781 176 481 198 923 508 803 215 199 467 887 262 646 419 780 362 305 536%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: corr_1260d (Cumula

Monthly Corr: -0.235 | Drift Check (R

Cumulative Return

300 000 000 000 000 031 996 193 956 364 717 059 030 036 345 467 498 327 588 019 124 832 485 129 925 441 359 970 304%

200 000 000 000 000 002 926 139 046 134 974 606 194 008 597 572 931 011 855 722 157 433 959 272 850 229 643 182 080%

100 000 000 000 000 001 463 069 523 067 487 303 097 004 298 786 465 505 927 861 078 716 979 636 425 114 821 591 040%

0%

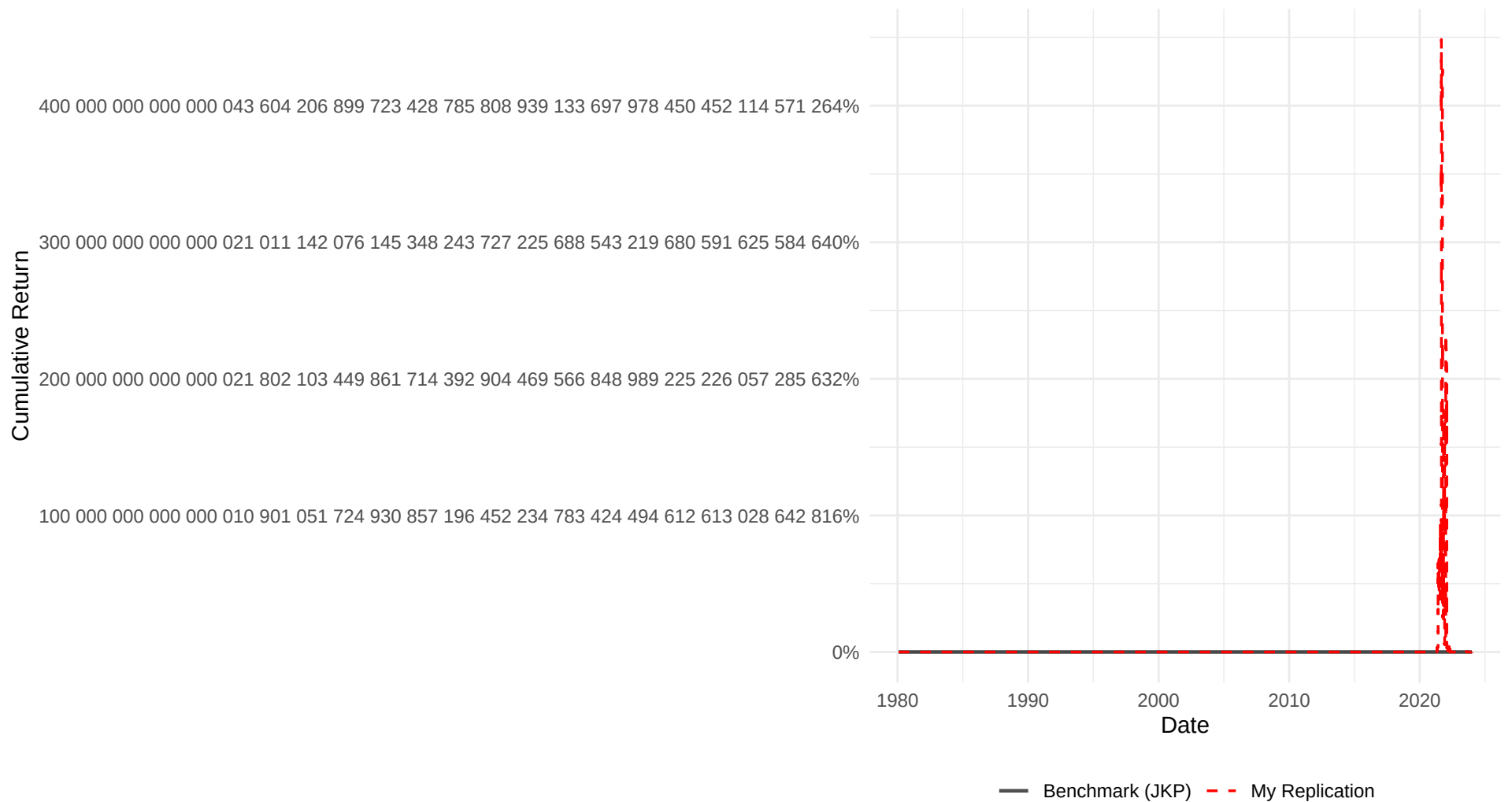
1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: eqpo_me (Cumulative Wealth)

Monthly Corr: -0.2331 | Drift Check (Red should overlap Black)



Factor: niq_at (Cumulative Wealth)

Monthly Corr: -0.232 | Drift Check (Red should overlap Black)

Cumulative Return

30 000 000 000 000 000 978 680 950 144 401 383 192 292 617 328 216 608 963 406 365 458 432%

19 999 999 999 999 999 654 735 515 678 371 196 634 479 565 751 161 864 557 154 294 300 672%

9 999 999 999 999 999 827 367 757 839 185 598 317 239 782 875 580 932 278 577 147 150 336%

0%

1980

1990

2000

2010

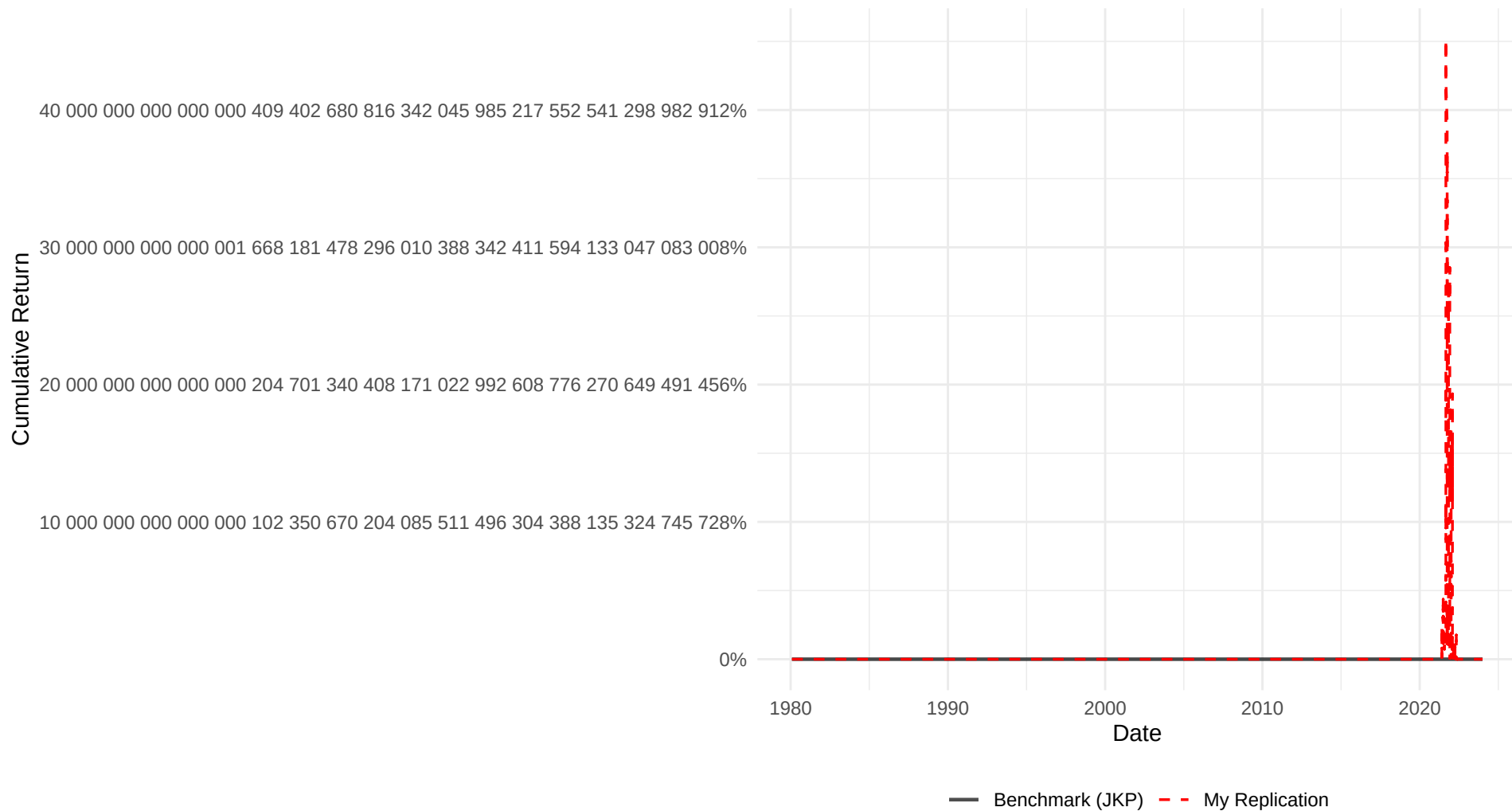
2020

Date

— Benchmark (JKP) - - My Replication

Factor: ocfq_saleq_std (Cumulative Wealth)

Monthly Corr: -0.2185 | Drift Check (Red should overlap Black)

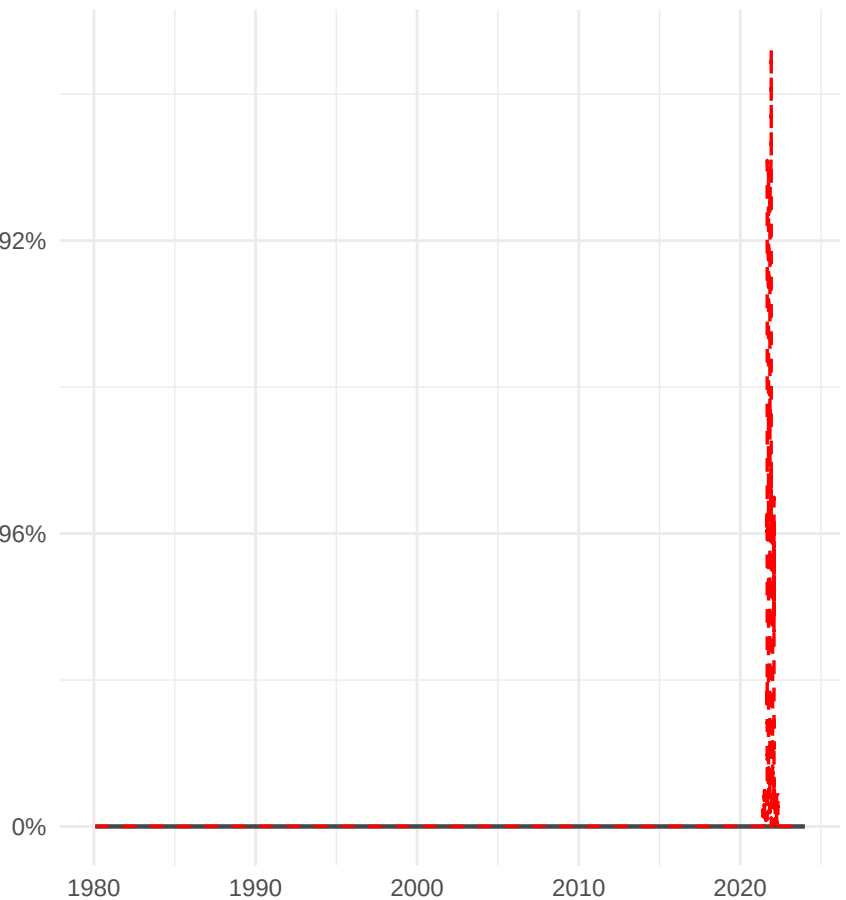


Factor: niq_be_chg1 (Cumulative Wealth)
Monthly Corr: -0.2158 | Drift Check (Red should overlap Black)

Cumulative Return

3 999 999 999 999 999 781 289 335 472 989 780 502 838 586 280 084 991 698 663 366 459 392%

1 999 999 999 999 999 890 644 667 736 494 890 251 419 293 140 042 495 849 331 683 229 696%



— Benchmark (JKP) - - My Replication

Factor: resff3_6_1 (Cumulative Wealth)

Monthly Corr: -0.2135 | Drift Check (Red should overlap Black)

Cumulative Return

1 499 999 999 999 999 995 443 087 580 909 022 353 359 765 988 994 871 928 763 187 200%

999 999 999 999 999 875 170 255 276 364 105 051 932 774 599 639 662 981 181 079 552%

499 999 999 999 999 937 585 127 638 182 052 525 966 387 299 819 831 490 590 539 776%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication

Factor: eqnpo_12m (C
Monthly Corr: -0.21 | Drift Che

Cumulative Return

749 999 999 999 999 945 688 905 802 977 242 987 289 481 237 198 023 593 891 341 842 771 664 580 074 072 827 271 145 586 688%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

249 999 999 999 999 981 896 301 934 325 747 662 429 827 079 066 007 864 630 447 280 923 888 193 358 024 275 757 048 528 896%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replicat

4 999 999 999 999 999 812 433 457 394 426 521 793 962 301 763 257 387 871 861 409 146 707 361 282 553 284 011 931 062 452 285 642 191 752 550 230 112 314 197 016 570

0 000 000 000 000 000 000 657 803 165 854 228 757 159 135 066 771 950 601 039 801 789 067 244 864 424 132 513 185 357 050 006 393 242 615 734 699 614 997 777 141 989 370

5 000 000 000 000 000 000 328 901 582 927 114 378 579 567 533 385 975 300 519 900 894 533 622 432 212 066 256 592 678 525 003 196 621 307 867 349 807 498 888 570 994 680

Factor: ni_be (Cumulative

Monthly Corr: -0.2071 | Drift Ch

Cumulative Return

7 500 000 000 000 000 144 404 587 876 337 340 200 347 535 861 064 688 817 703 952 379 957 494 785 020 133 319 695 663 104%

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

2 499 999 999 999 999 746 592 963 569 934 854 484 566 405 160 230 136 238 010 732 393 213 529 408 919 252 752 624 779 264%

0%

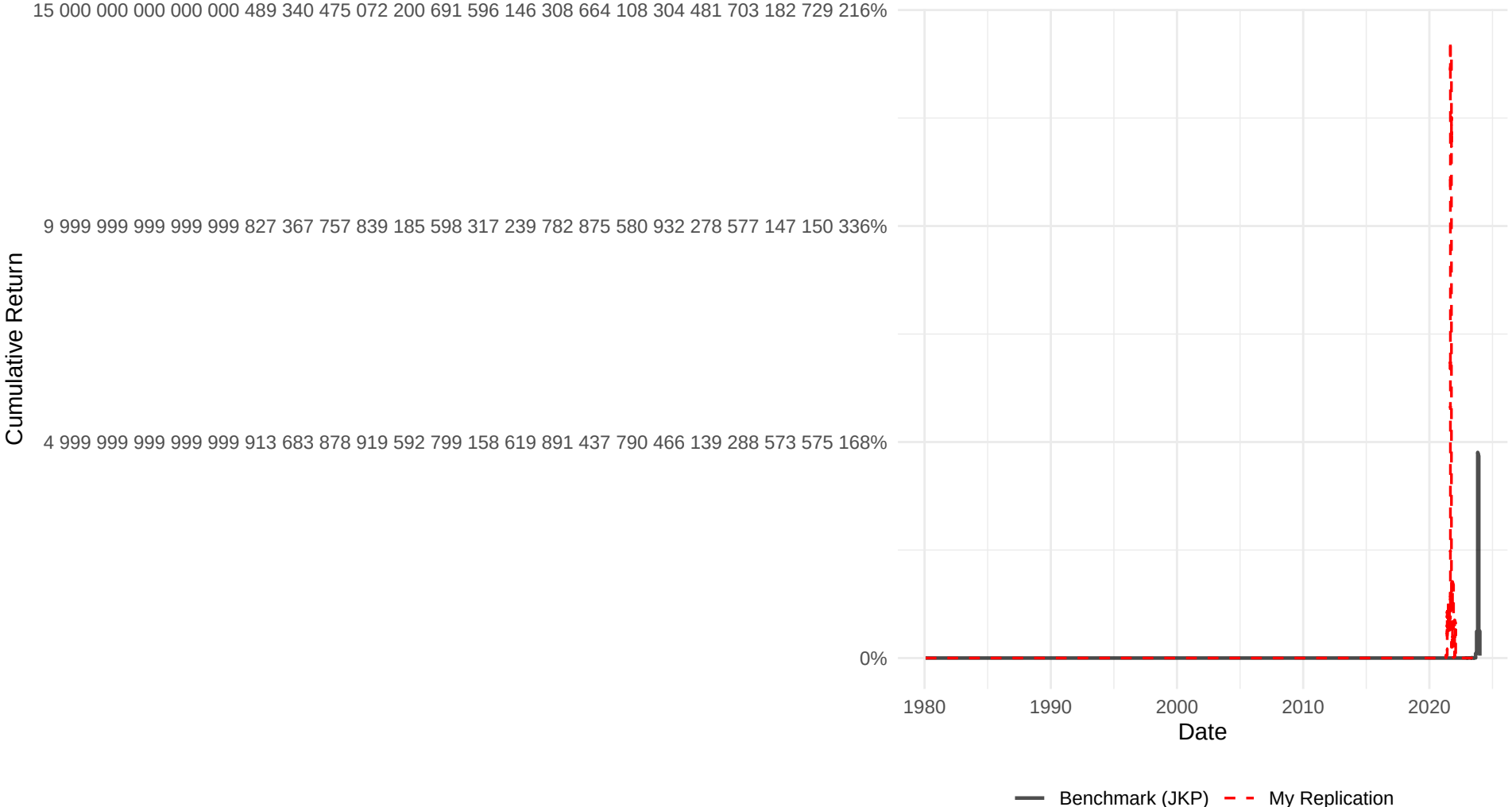
1980 1990 2000 2010 2020

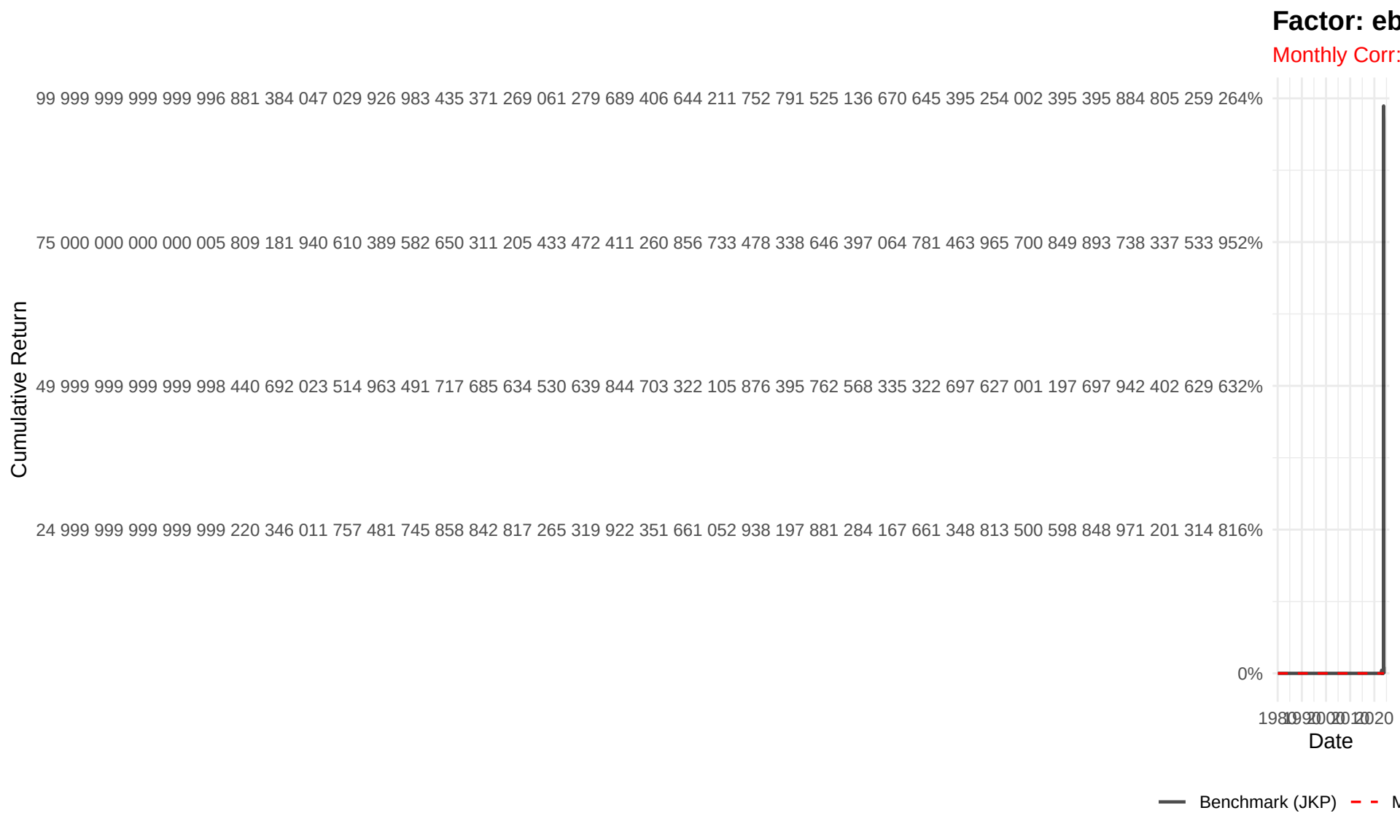
Date

— Benchmark (JKP) - - My Replicati

Factor: eqnetis_at (Cumulative Wealth)

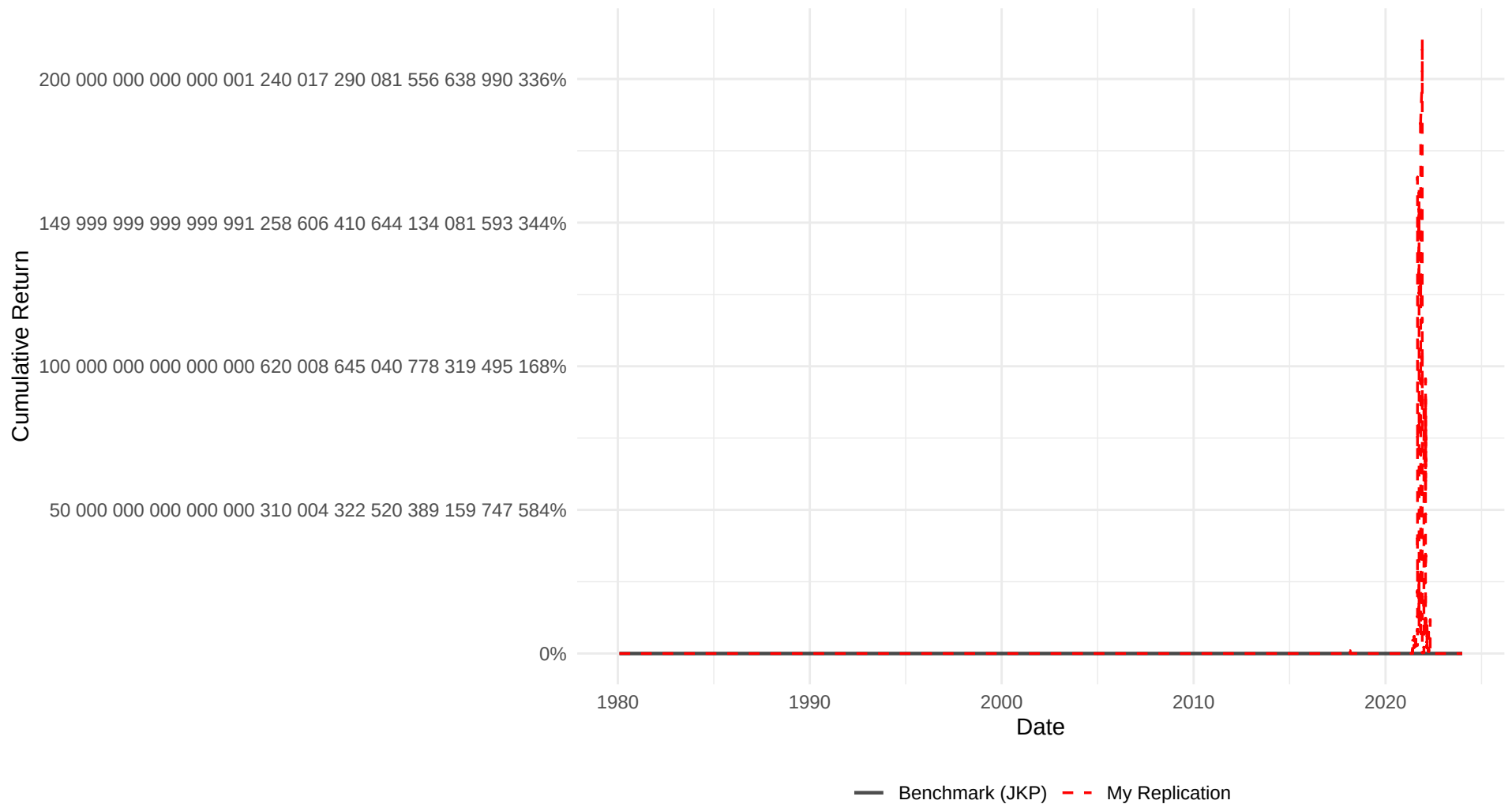
Monthly Corr: -0.2069 | Drift Check (Red should overlap Bla





Monthly Corr: -0.2044 | Drift Check (Red should overlap Black)

Monthly Corr: -0.2044 | Drift Check (Red should overlap Black)





Factor: capx_gr3 (Cumulative W

Monthly Corr: -0.2024 | Drift Check (Red sl

Cumulative Return

1 999 999 999 999 999 842 563 759 791 331 565 483 871 006 498 118 367 703 619 996 448 246 128 296 859 795 456%

1 499 999 999 999 999 776 610 528 174 941 487 414 985 227 189 918 343 458 819 901 935 635 484 968 333 869 056%

999 999 999 999 999 921 281 879 895 665 782 741 935 503 249 059 183 851 809 998 224 123 064 148 429 897 728%

499 999 999 999 999 960 640 939 947 832 891 370 967 751 624 529 591 925 904 999 112 061 532 074 214 948 864%

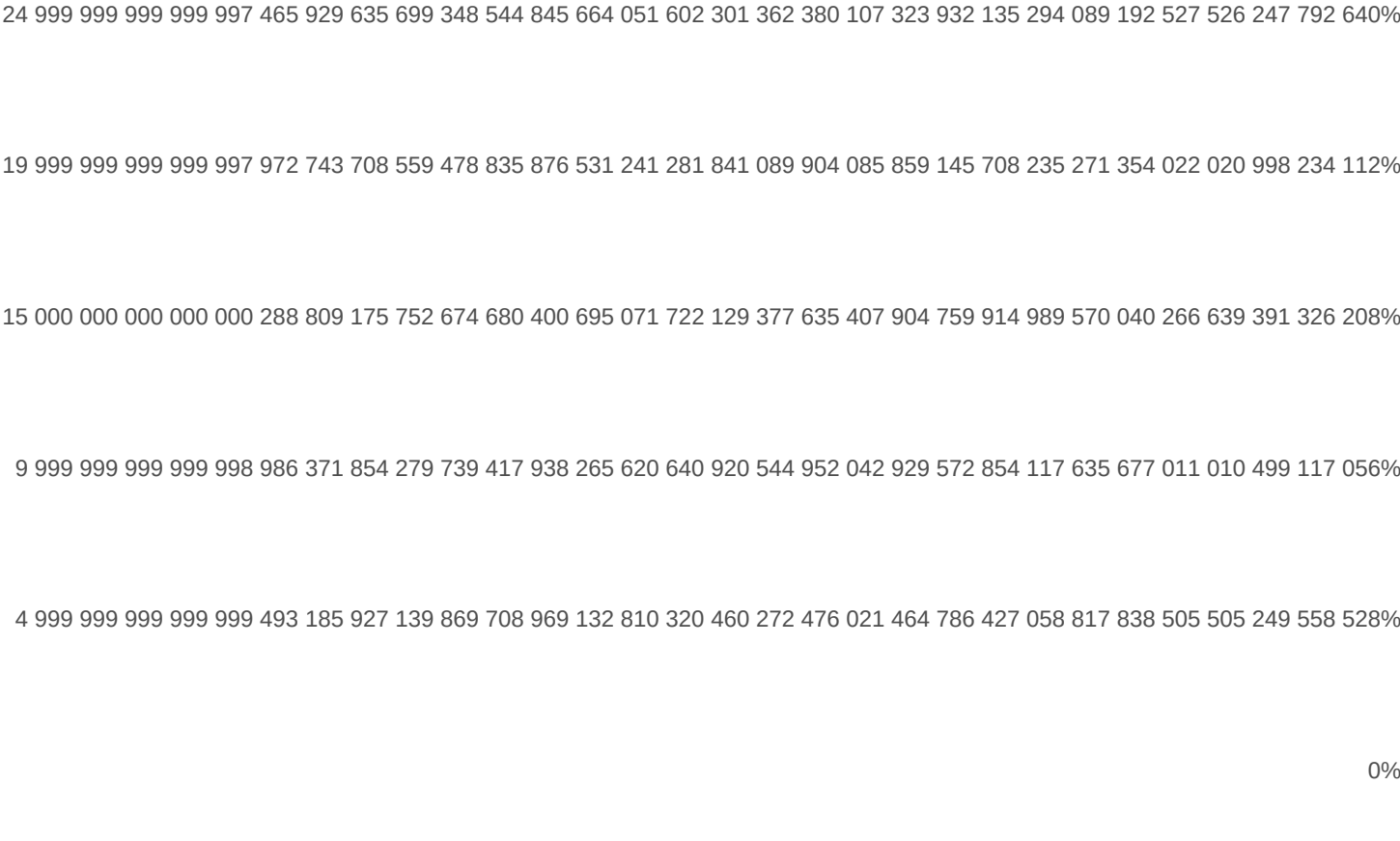
0%

1980 1990 2000 2010 2020
Date

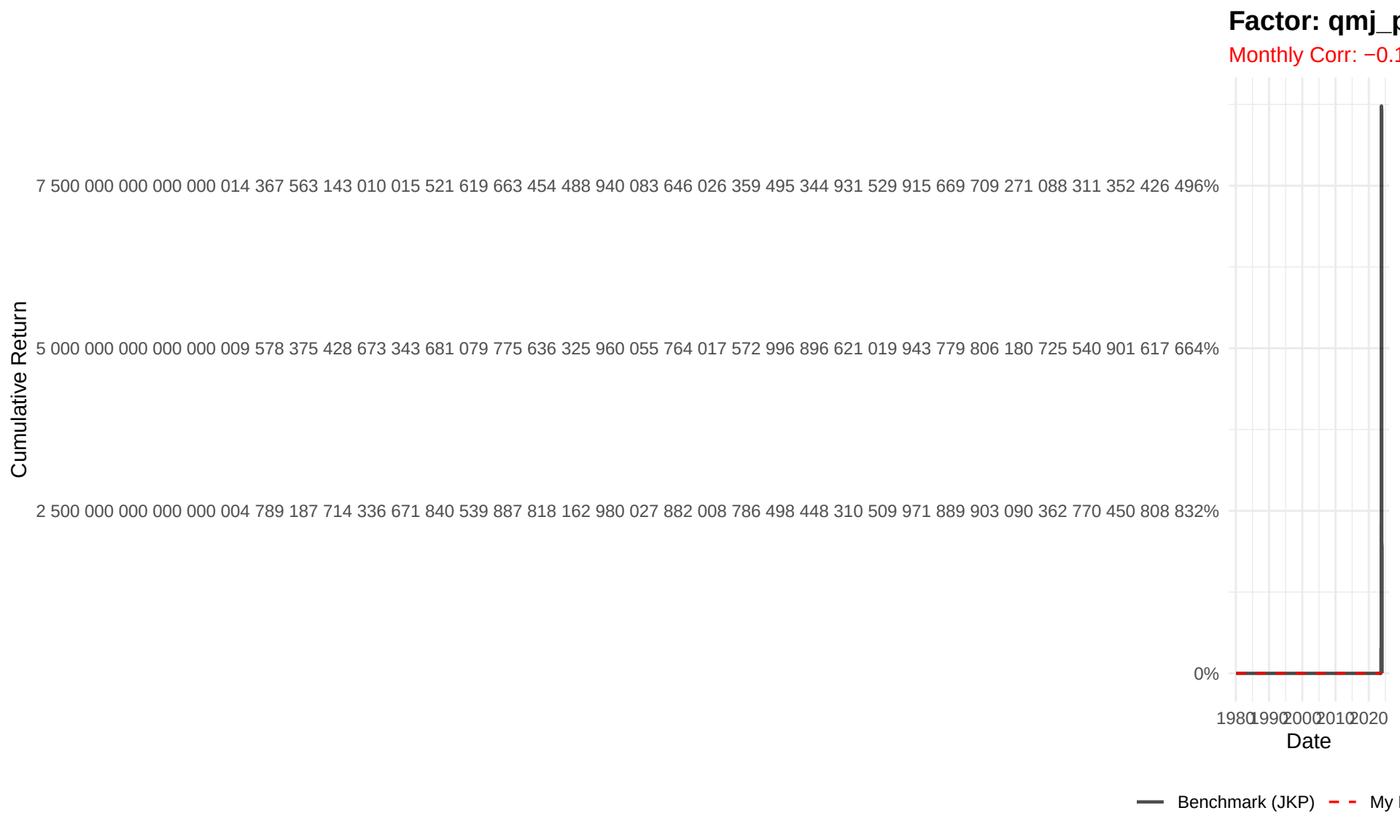
— Benchmark (JKP) - - My Replication

Factor: aliq_at (Cumula
Monthly Corr: -0.2019 | Drift CH

Cumulative Return



— Benchmark (JKP) - - My Replication



Factor: o_score (Cumulative)

Monthly Corr: -0.1909 | Drift Check

Cumulative Return

199 999 999 999 999 998 950 733 150 383 609 864 631 589 220 901 364 351 243 883 389 463 816 617 076 615 690 273 685 504%

149 999 999 999 999 997 010 885 085 447 095 443 474 121 906 761 001 073 503 105 779 590 434 965 348 187 352 193 826 816%

99 999 999 999 999 999 475 366 575 191 804 932 315 794 610 450 682 175 621 941 694 731 908 308 538 307 845 136 842 752%

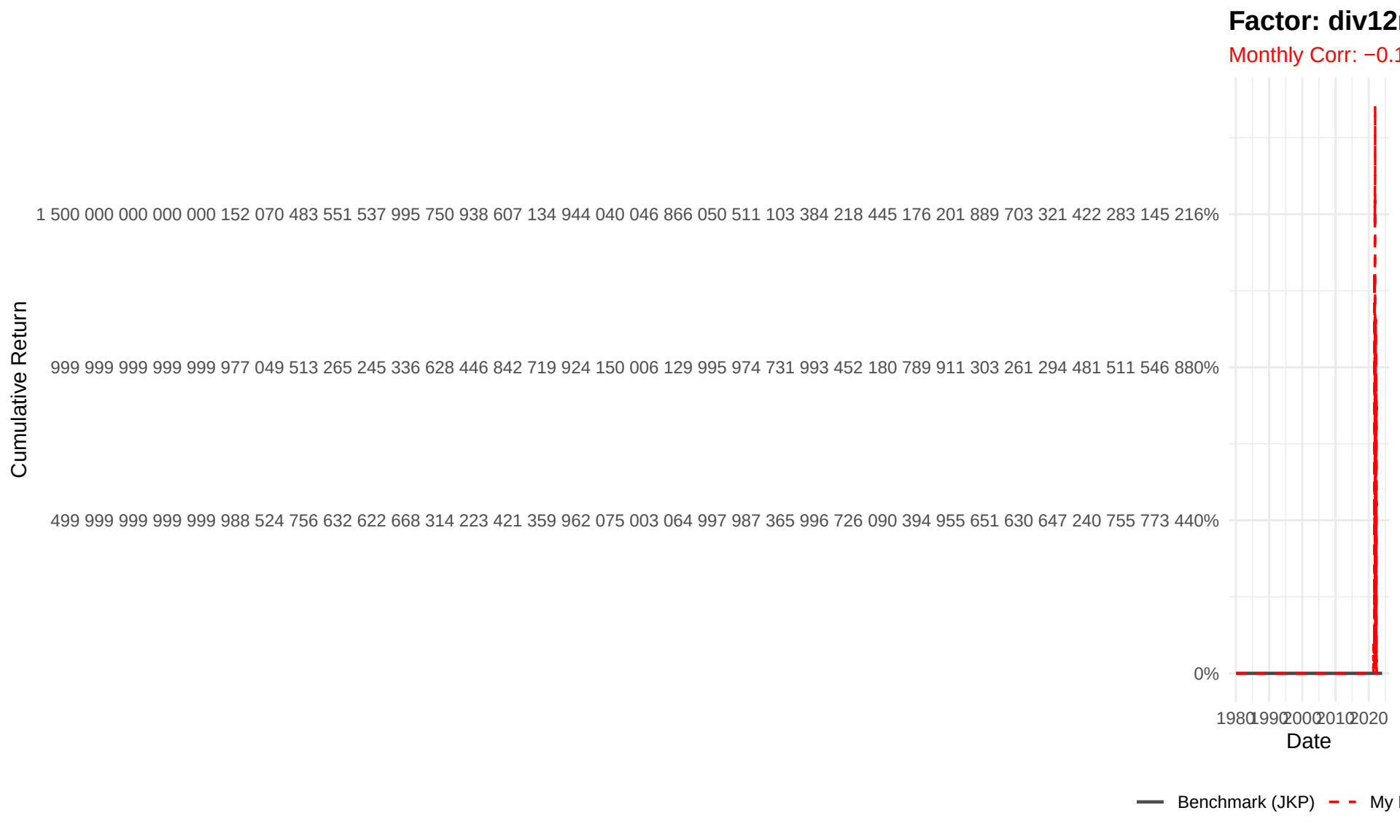
49 999 999 999 999 999 737 683 287 595 902 466 157 897 305 225 341 087 810 970 847 365 954 154 269 153 922 568 421 376%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replication



Factor: resff3_12_1 (Cumulative Wealth)

Monthly Corr: -0.188 | Drift Check (Red should overlap Black)

Cumulative Return

74 999 999 999 999 999 772 154 379 045 451 117 667 988 299 449 743 596 438 159 360%

49 999 999 999 999 996 042 109 072 147 741 062 289 896 281 101 175 367 271 448 576%

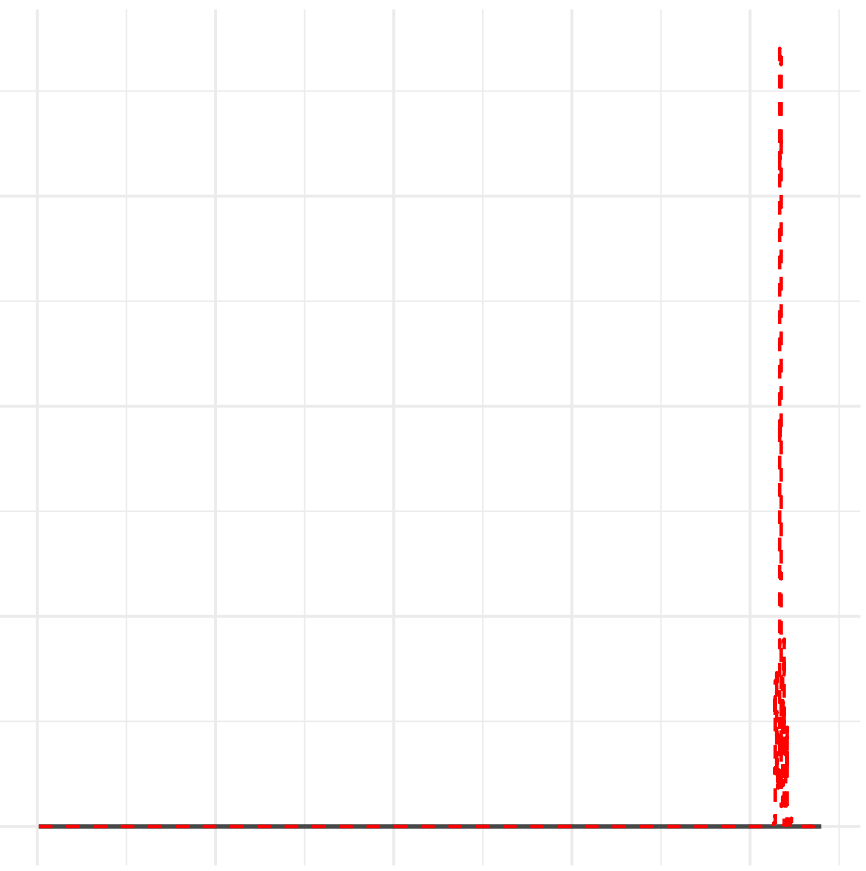
24 999 999 999 999 998 021 054 536 073 870 531 144 948 140 550 587 683 635 724 288%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication



Factor: noa_gr1a (Cumulative Return)
Monthly Corr: -0.1868 | Drift Coefficient: 0.0001

Cumulative Return

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

749 999 999 999 999 945 688 905 802 977 242 987 289 481 237 198 023 593 891 341 842 771 664 580 074 072 827 271 145 586 688%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

249 999 999 999 999 981 896 301 934 325 747 662 429 827 079 066 007 864 630 447 280 923 888 193 358 024 275 757 048 528 896%



— Benchmark (JKP) - - - My Replication

Factor: netis_at (Cumulative Wealth)

Monthly Corr: -0.1861 | Drift Check (Red should overlap)

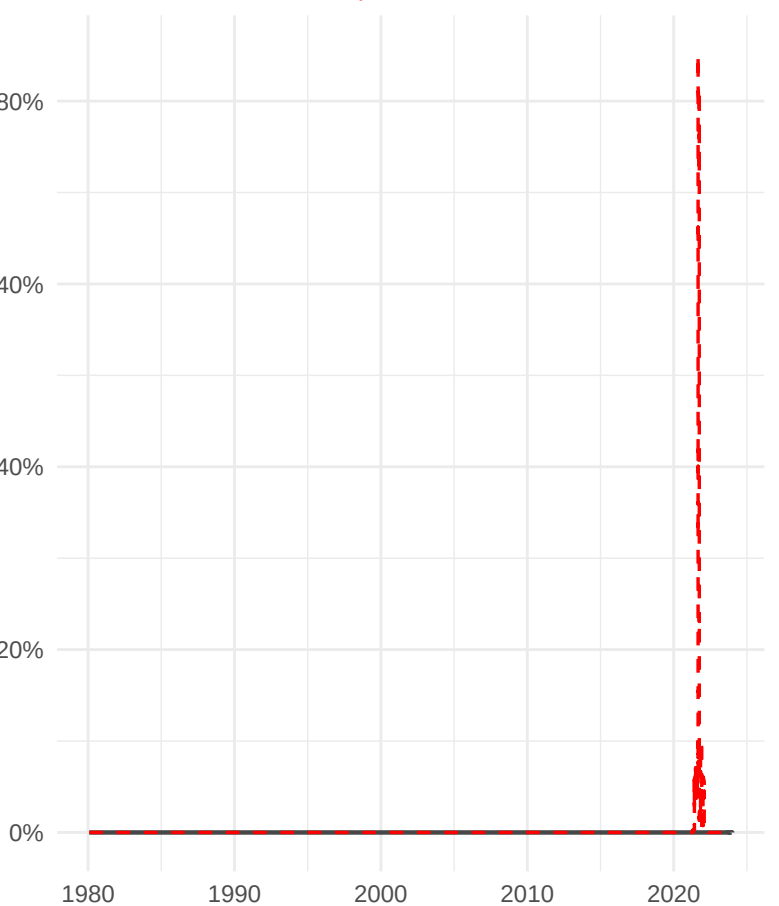
Cumulative Return

199 999 999 999 999 983 856 376 458 988 069 858 064 734 338 923 120 718 728 859 587 423 764 480%

149 999 999 999 999 957 242 371 526 923 275 226 381 610 211 186 156 866 671 862 246 895 779 840%

99 999 999 999 999 991 928 188 229 494 034 929 032 367 169 461 560 359 364 429 793 711 882 240%

49 999 999 999 999 995 964 094 114 747 017 464 516 183 584 730 780 179 682 214 896 855 941 120%



— Benchmark (JKP) - - My Replication

Factor: capx_gr2 (Cumulative

Monthly Corr: -0.1826 | Drift Check (Re

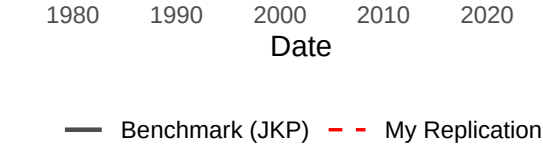
Cumulative Return

7 999 999 999 999 998 736 696 292 487 286 226 760 809 171 447 512 149 984 007 091 863 281 059 881 504 340 967 424%

5 999 999 999 999 999 915 240 512 714 285 143 499 951 361 370 262 294 044 393 940 418 759 114 306 443 783 700 480%

3 999 999 999 999 999 368 348 146 243 643 113 380 404 585 723 756 074 992 003 545 931 640 529 940 752 170 483 712%

1 999 999 999 999 999 684 174 073 121 821 556 690 202 292 861 878 037 496 001 772 965 820 264 970 376 085 241 856%



Factor: ppeinv_gr1a (Cu

Monthly Corr: -0.1797 | Drift Ch

Cumulative Return

3 999 999 999 999 999 865 936 450 861 855 600 199 300 744 370 480 502 011 918 698 389 236 719 021 749 516 922 746 044 416%

2 999 999 999 999 999 786 374 126 000 575 103 056 144 518 230 313 591 495 980 054 391 887 925 946 529 340 809 331 867 648%

1 999 999 999 999 999 932 968 225 430 927 800 099 650 372 185 240 251 005 959 349 194 618 359 510 874 758 461 373 022 208%

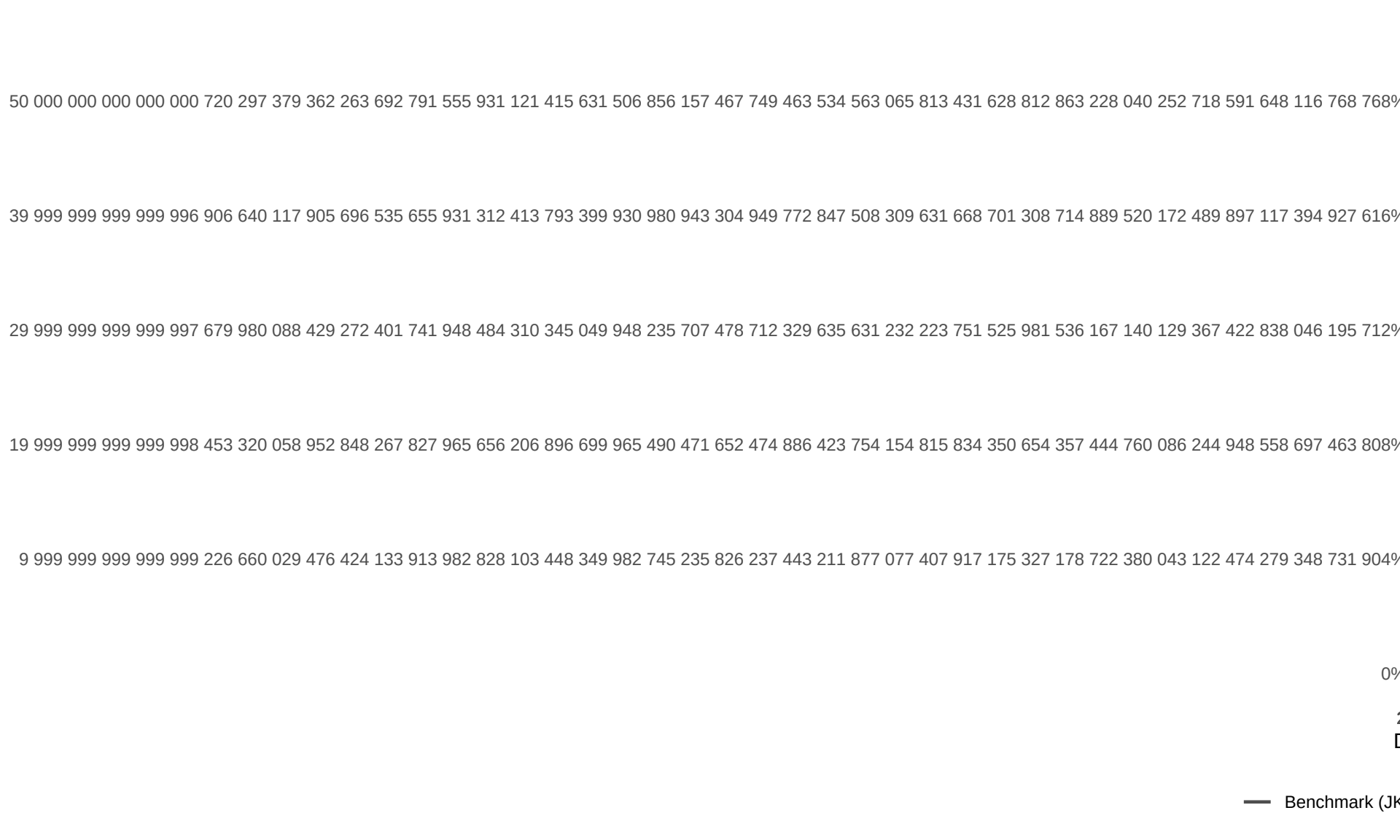
999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati



Factor: chcscho_12m (Cumulative Wea

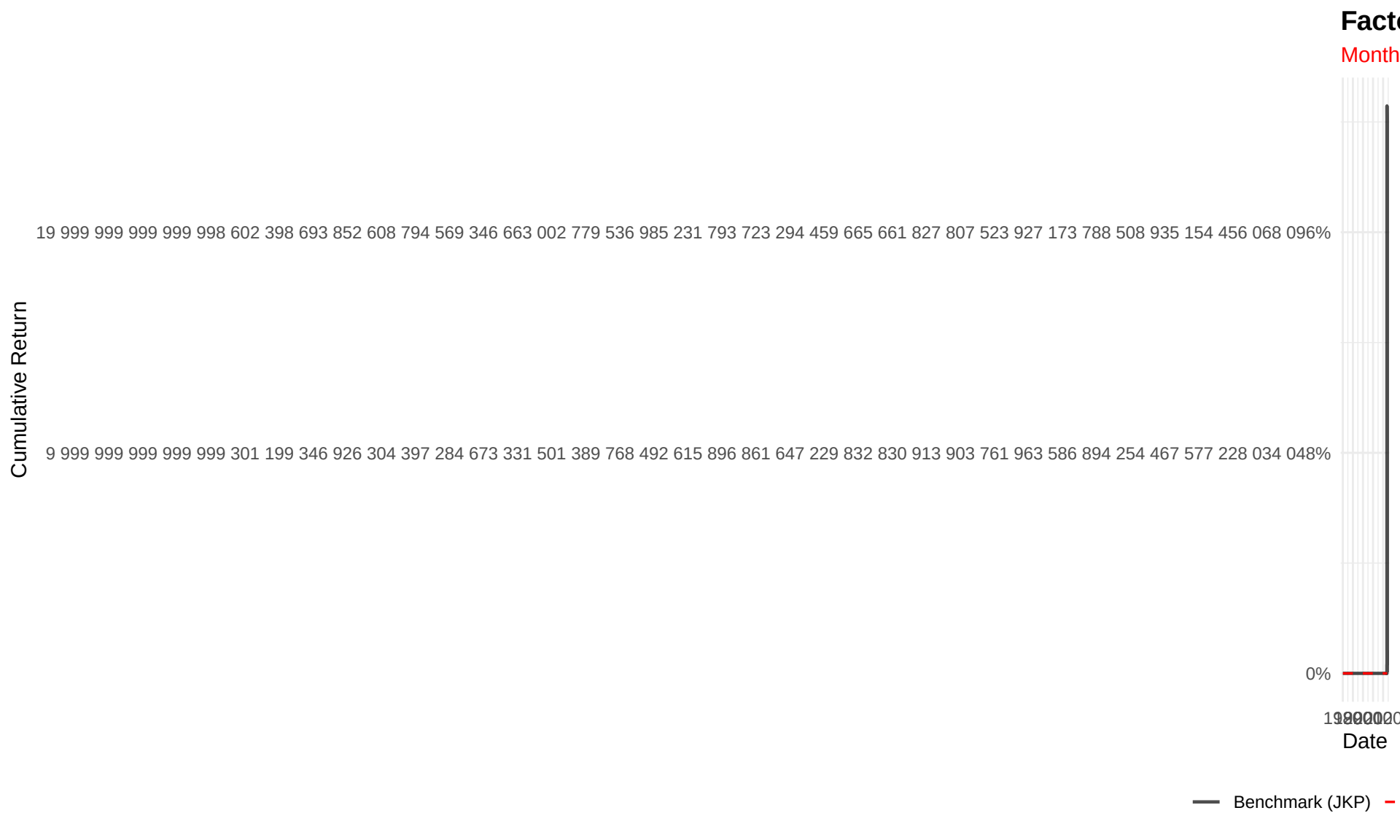
Monthly Corr: -0.1638 | Drift Check (Red should ove

Cumulative Return

125 000 000 000 000 015 925 879 588 606 872 920 781 758 094 730 599 150 317 252 580 328 683 339 776%
100 000 000 000 000 007 719 022 282 576 153 725 556 774 937 218 346 187 371 917 708 691 719 061 504%
75 000 000 000 000 012 066 368 447 318 796 058 003 370 626 121 426 056 631 293 725 982 823 809 024%
50 000 000 000 000 003 859 511 141 288 076 862 778 387 468 609 173 093 685 958 854 345 859 530 752%
25 000 000 000 000 001 929 755 570 644 038 431 389 193 734 304 586 546 842 979 427 172 929 765 376%
0%



— Benchmark (JKP) - - My Replication



Factor: capx_gr1 (Cumulative Wea

Monthly Corr: -0.1591 | Drift Check (Red sho

Cumulative Return

39 999 999 999 999 998 694 240 300 026 382 076 890 985 614 426 599 919 653 064 098 474 532 012 887 638 016%

29 999 999 999 999 997 375 175 667 698 580 515 513 270 028 262 599 434 757 062 208 222 319 146 317 119 488%

19 999 999 999 999 999 347 120 150 013 191 038 445 492 807 213 299 959 826 532 049 237 266 006 443 819 008%

9 999 999 999 999 999 673 560 075 006 595 519 222 746 403 606 649 979 913 266 024 618 633 003 221 909 504%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: sale_gr3 (Cumulative)

Monthly Corr: -0.1584 | Drift

Cumulative Return

3 000 000 000 000 000 000 245 923 980 161 173 753 643 441 864 983 543 725 788 645 306 033 648 914 478 126 627 340 737 100 906 496%

2 000 000 000 000 000 000 086 754 593 949 238 372 146 580 586 649 903 878 623 583 547 578 672 233 625 779 362 221 882 647 511 040%

1 000 000 000 000 000 000 043 377 296 974 619 186 073 290 293 324 951 939 311 791 773 789 336 116 812 889 681 110 941 323 755 520%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: inv_gr1 (Cumulat

Monthly Corr: -0.1544 | Drift Chec

Cumulative Return

149 999 999 999 999 978 010 885 085 447 095 443 474 121 906 761 001 073 503 105 779 590 434 965 348 187 352 193 826 816%

99 999 999 999 999 999 999 475 366 575 191 804 932 315 794 610 450 682 175 621 941 694 731 908 308 538 307 845 136 842 752%

49 999 999 999 999 999 999 737 683 287 595 902 466 157 897 305 225 341 087 810 970 847 365 954 154 269 153 922 568 421 376%

0%

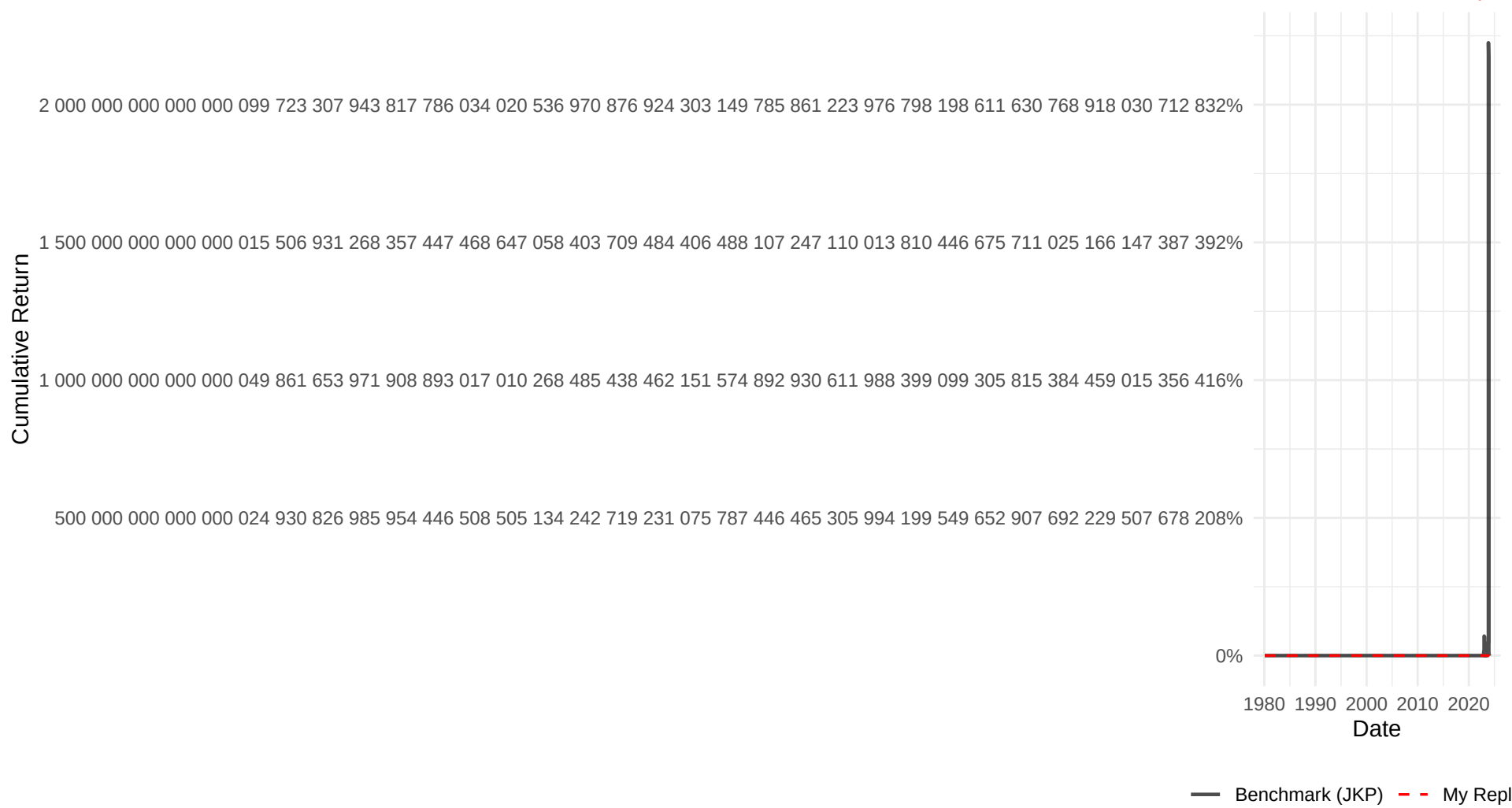
1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: ope_be (Cu

Monthly Corr: -0.152 | Dr



Factor: seas_16_20na (Cumulative Wealth)

Monthly Corr: -0.1512 | Drift Check (Red should overlap Black)

Cumulative Return

100 000 000 000 000 000 620 008 645 040 778 319 495 168%

74 999 999 999 999 995 629 303 205 322 067 040 796 672%

50 000 000 000 000 000 310 004 322 520 389 159 747 584%

25 000 000 000 000 000 155 002 161 260 194 579 873 792%

0%

1980

1990

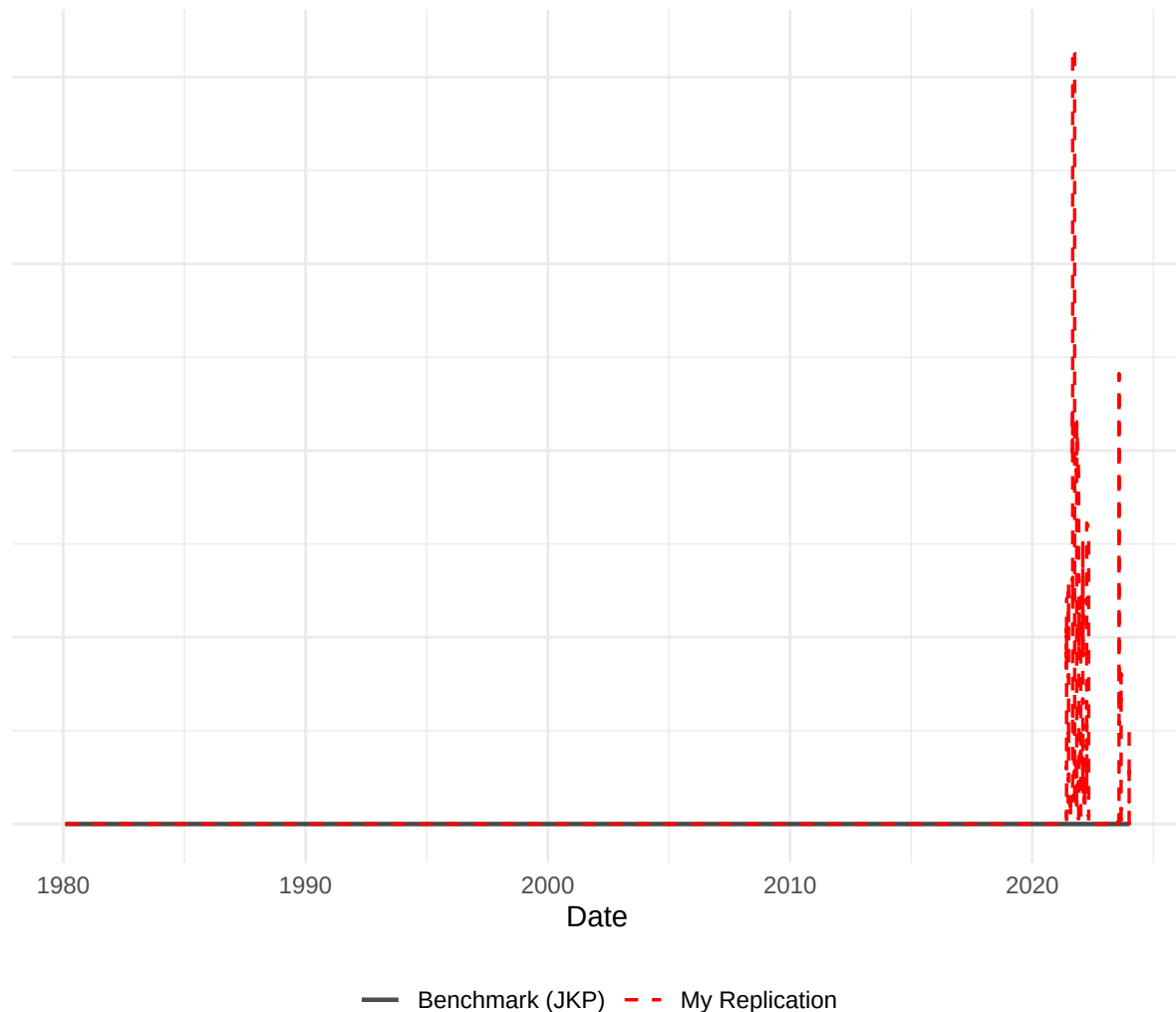
2000

2010

2020

Date

— Benchmark (JKP) - - My Replication



Factor: sale_gr1 (Cumulative)

Monthly Corr: -0.1511 | Drift CH

Cumulative Return

19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%

9 999 999 999 999 998 986 371 854 279 739 417 938 265 620 640 920 544 952 042 929 572 854 117 635 677 011 010 499 117 056%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: seas_6_10na (Cumulative
Monthly Corr: -0.1507 | Drift Check (Red show

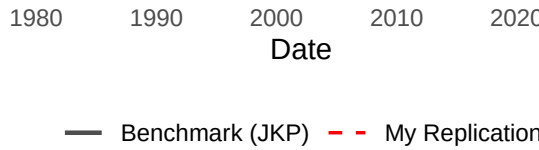
Cumulative Return

59 999 999 999 999 994 750 351 335 397 161 031 026 540 056 525 198 869 514 124 416 444 638 292 634 238 976%

39 999 999 999 999 998 694 240 300 026 382 076 890 985 614 426 599 919 653 064 098 474 532 012 887 638 016%

19 999 999 999 999 999 347 120 150 013 191 038 445 492 807 213 299 959 826 532 049 237 266 006 443 819 008%

0%



Factor: nncoa_gr1a (

Monthly Corr: -0.1491 | Drift

Cumulative Return

1 499 999 999 999 999 891 377 811 605 954 485 974 578 962 474 396 047 187 782 683 685 543 329 160 148 145 654 542 291 173 376%

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: ret_12_1

Monthly Corr: -0.1483

Cumulative Return

4 000 000 000 000 000 355 011 898 272 899 032 825 263 609 449 105 948 552 273 556 880 920 063 696 640 013 885 161 030 775 124 000 768%

3 000 000 000 000 000 144 842 117 940 566 207 686 481 337 910 359 529 749 054 740 219 931 327 694 241 734 805 189 255 256 017 469 440%

2 000 000 000 000 000 177 505 949 136 449 516 412 631 804 724 552 974 276 136 778 440 460 031 848 320 006 942 580 515 387 562 000 384%

1 000 000 000 000 000 088 752 974 568 224 758 206 315 902 362 276 487 138 068 389 220 230 015 924 160 003 471 290 257 693 781 000 192%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Re

Factor: niq_su (Cumulative Wealth)

Monthly Corr: -0.1478 | Drift Check (Red should overlap Black)

Cumulative Return

999 999 999 999 999 993 220 948 674 361 627 976 461 708 441 944 064%

499 999 999 999 999 996 610 474 337 180 813 988 230 854 220 972 032%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication

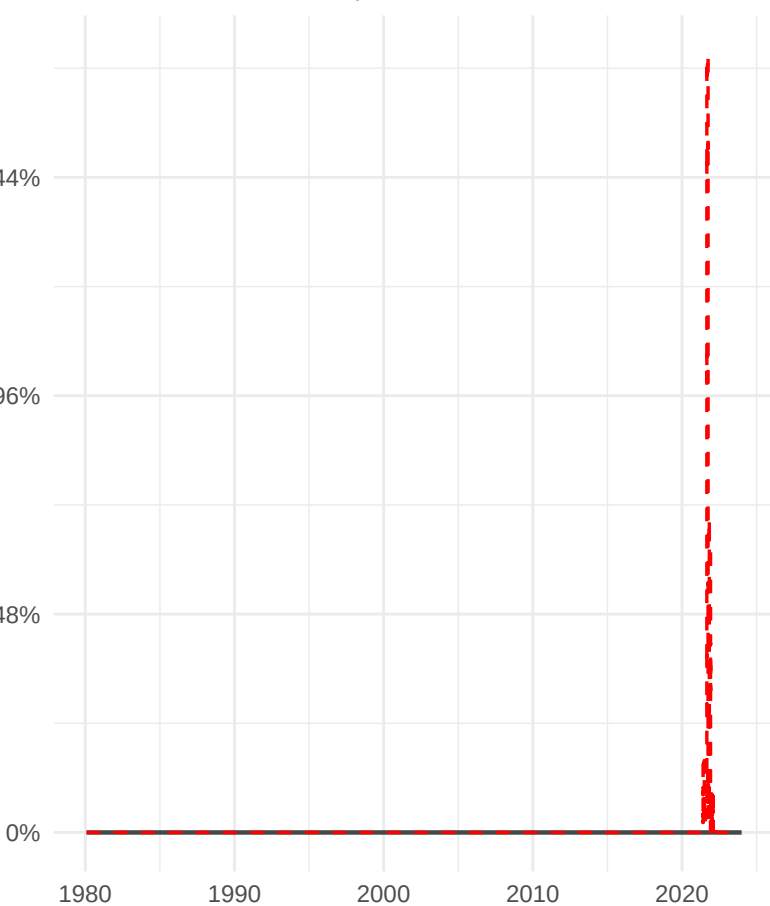
Factor: emp_gr1 (Cumulative Wealth)
Monthly Corr: -0.1468 | Drift Check (Red should overlap)

Cumulative Return

59 999 999 999 999 995 156 912 937 696 420 957 419 420 301 676 936 215 618 657 876 227 129 344%

39 999 999 999 999 996 771 275 291 797 613 971 612 946 867 784 624 143 745 771 917 484 752 896%

19 999 999 999 999 998 385 637 645 898 806 985 806 473 433 892 312 071 872 885 958 742 376 448%



— Benchmark (JKP) - - - My Replication

Factor: coa_gr1a (Cum

Monthly Corr: -0.1458 | Drift C

Cumulative Return

599 999 999 999 999 910 234 288 947 455 316 220 403 190 986 283 255 733 805 079 607 961 106 048 276 224 658 651 664 613 376%

399 999 999 999 999 959 454 874 171 189 576 717 530 624 825 636 821 798 081 717 182 914 164 705 427 080 440 419 964 682 240%

199 999 999 999 999 979 727 437 085 594 788 358 765 312 412 818 410 899 040 858 591 457 082 352 713 540 220 209 982 341 120%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replicat

Factor: niq_at_chg1 (Cumulative Wealth)
Monthly Corr: -0.1384 | Drift Check (Red should overlap Black)

Cumulative Return

19 999 999 999 999 998 987 742 705 941 480 377 339 272 900 220 268 201 461 678 080%

14 999 999 999 999 999 954 430 875 809 090 223 533 597 659 889 948 719 287 631 872%

9 999 999 999 999 999 493 871 352 970 740 188 669 636 450 110 134 100 730 839 040%

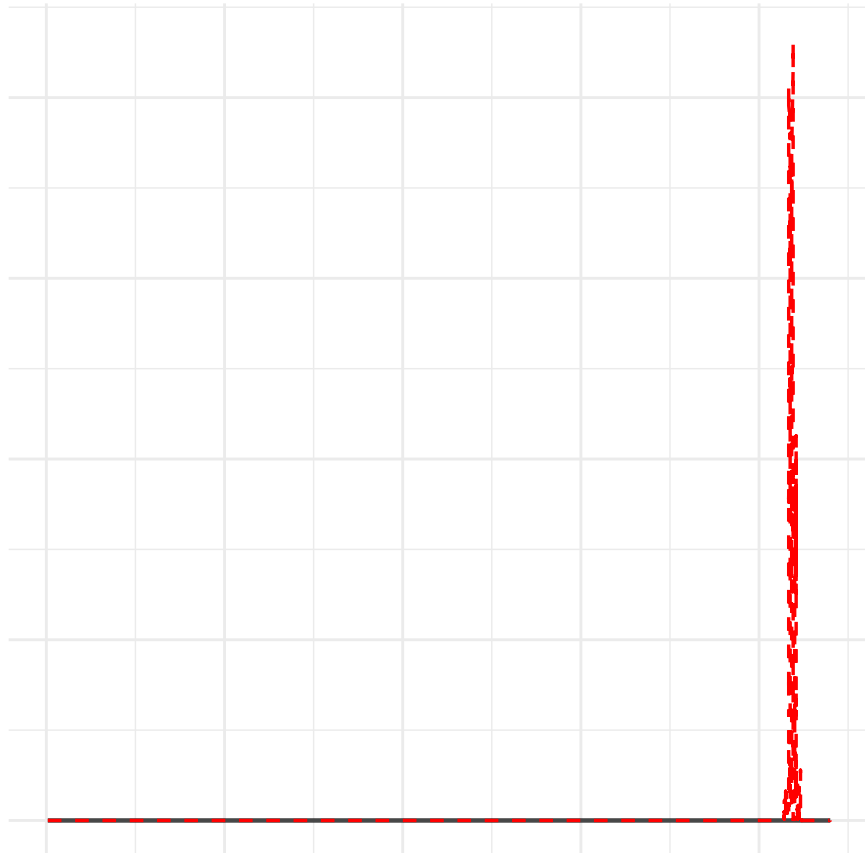
4 999 999 999 999 999 746 935 676 485 370 094 334 818 225 055 067 050 365 419 520%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication



Factor: ret_9_1 (

Monthly Corr: -0.1382

Cumulative Return

4 000 000 000 000 000 355 011 898 272 899 032 825 263 609 449 105 948 552 273 556 880 920 063 696 640 013 885 161 030 775 124 000 768%

3 000 000 000 000 000 144 842 117 940 566 207 686 481 337 910 359 529 749 054 740 219 931 327 694 241 734 805 189 255 256 017 469 440%

2 000 000 000 000 000 177 505 949 136 449 516 412 631 804 724 552 974 276 136 778 440 460 031 848 320 006 942 580 515 387 562 000 384%

1 000 000 000 000 000 088 752 974 568 224 758 206 315 902 362 276 487 138 068 389 220 230 015 924 160 003 471 290 257 693 781 000 192%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Re

Factor: ret_3_1

Monthly Corr: -0.136

Cumulative Return

40 000 000 000 000 000 636 115 644 390 396 721 873 443 234 255 781 125 559 125 310 230 991 355 088 681 524 243 253 879 943 427 260 416%

30 000 000 000 000 000 002 419 755 625 518 526 612 324 544 332 515 354 750 811 750 821 725 383 037 568 323 552 921 344 695 162 778 943 488%

20 000 000 000 000 000 000 318 057 822 195 198 360 936 721 617 127 890 562 779 562 655 115 495 677 544 340 762 121 626 939 971 713 630 208%

10 000 000 000 000 000 000 159 028 911 097 599 180 468 360 808 563 945 281 389 781 327 557 747 838 772 170 381 060 813 469 985 856 815 104%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My R

Factor: ni_me (Cumulative)

Monthly Corr: -0.1362 | Drift Change

Cumulative Return

1 199 999 999 999 999 824 087 080 683 576 763 547 792 975 254 088 008 588 024 846 236 723 479 722 785 498 817 550 614 528%

799 999 999 999 999 995 802 932 601 534 439 458 526 356 883 605 457 404 975 533 557 855 266 468 306 462 761 094 742 016%

399 999 999 999 999 997 901 466 300 767 219 729 263 178 441 802 728 702 487 766 778 927 633 234 153 231 380 547 371 008%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: Inoa_gr1a (Cumulative)

Monthly Corr: -0.136 | Drift Check: 0.000

Cumulative Return

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

3 999 999 999 999 999 865 936 450 861 855 600 199 300 744 370 480 502 011 918 698 389 236 719 021 749 516 922 746 044 416%

2 999 999 999 999 999 786 374 126 000 575 103 056 144 518 230 313 591 495 980 054 391 887 925 946 529 340 809 331 867 648%

1 999 999 999 999 999 932 968 225 430 927 800 099 650 372 185 240 251 005 959 349 194 618 359 510 874 758 461 373 022 208%

999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: at_gr1 (Cumula

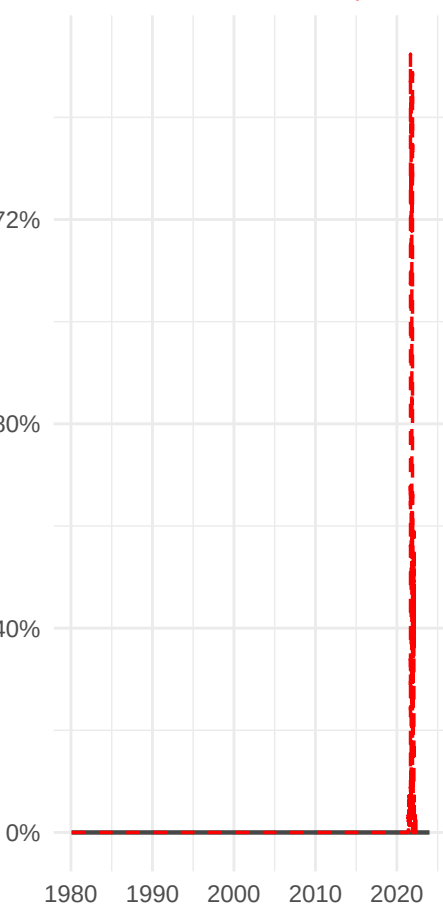
Monthly Corr: -0.1292 | Drift Ch

Cumulative Return

74 999 999 999 999 988 779 286 118 431 914 527 550 398 873 285 406 966 725 634 950 995 138 256 034 528 082 331 458 076 672%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

24 999 999 999 999 997 465 929 635 699 348 544 845 664 051 602 301 362 380 107 323 932 135 294 089 192 527 526 247 792 640%



— Benchmark (JKP) - - My Replication

Factor: ret_6_1

Monthly Corr: -0.12

Cumulative Return

399 999 999 999 999 999 990 819 805 306 098 134 651 378 737 087 969 660 002 451 998 389 892 797 380 872 315 964 521 304 517 792 604 618 752%

199 999 999 999 999 999 995 409 902 653 049 067 325 689 368 543 984 830 001 225 999 194 946 398 690 436 157 982 260 652 258 896 302 309 376%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My R

Factor: saleq_gr1 (Cumulative Wealth)

Monthly Corr: -0.1248 | Drift Check (Red should overlap Black)

Cumulative Return

749 999 999 999 999 997 721 543 790 454 511 176 679 882 994 497 435 964 381 593 600%

499 999 999 999 999 937 585 127 638 182 052 525 966 387 299 819 831 490 590 539 776%

249 999 999 999 999 968 792 563 819 091 026 262 983 193 649 909 915 745 295 269 888%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication

Factor: seas_1_1

Monthly Corr: -0.1243

Cumulative Return

600 000 000 000 000 028 968 423 588 113 241 537 296 267 582 071 905 949 810 948 043 986 265 538 848 346 961 037 851 051 203 493 888%

399 999 999 999 999 999 076 148 098 057 483 202 786 450 191 969 615 355 682 227 455 864 390 346 192 518 705 911 647 729 914 740 736%

199 999 999 999 999 999 538 074 049 028 741 601 393 225 095 984 807 677 841 113 727 932 195 173 096 259 352 955 823 864 957 370 368%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Rep

Factor: col_gr1a (Cum

Monthly Corr: -0.1189 | Drift C

Cumulative Return

799 999 999 999 999 918 909 748 342 379 153 435 061 249 651 273 643 596 163 434 365 828 329 410 854 160 880 839 929 364 480%

599 999 999 999 999 910 234 288 947 455 316 220 403 190 986 283 255 733 805 079 607 961 106 048 276 224 658 651 664 613 376%

399 999 999 999 999 959 454 874 171 189 576 717 530 624 825 636 821 798 081 717 182 914 164 705 427 080 440 419 964 682 240%

199 999 999 999 999 979 727 437 085 594 788 358 765 312 412 818 410 899 040 858 591 457 082 352 713 540 220 209 982 341 120%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicat

Factor: inv_gr1a (Cumulat

Monthly Corr: -0.1187 | Drift Check

Cumulative Return

30 000 000 000 000 004 083 042 928 025 663 870 694 652 384 918 209 090 672 543 860 921 057 992 053 347 236 643 340 288%

19 999 999 999 999 999 188 334 489 127 007 254 629 839 921 792 969 028 793 394 780 196 134 078 459 019 088 510 320 640%

9 999 999 999 999 999 594 167 244 563 503 627 314 919 960 896 484 514 396 697 390 098 067 039 229 509 544 255 160 320%

0%

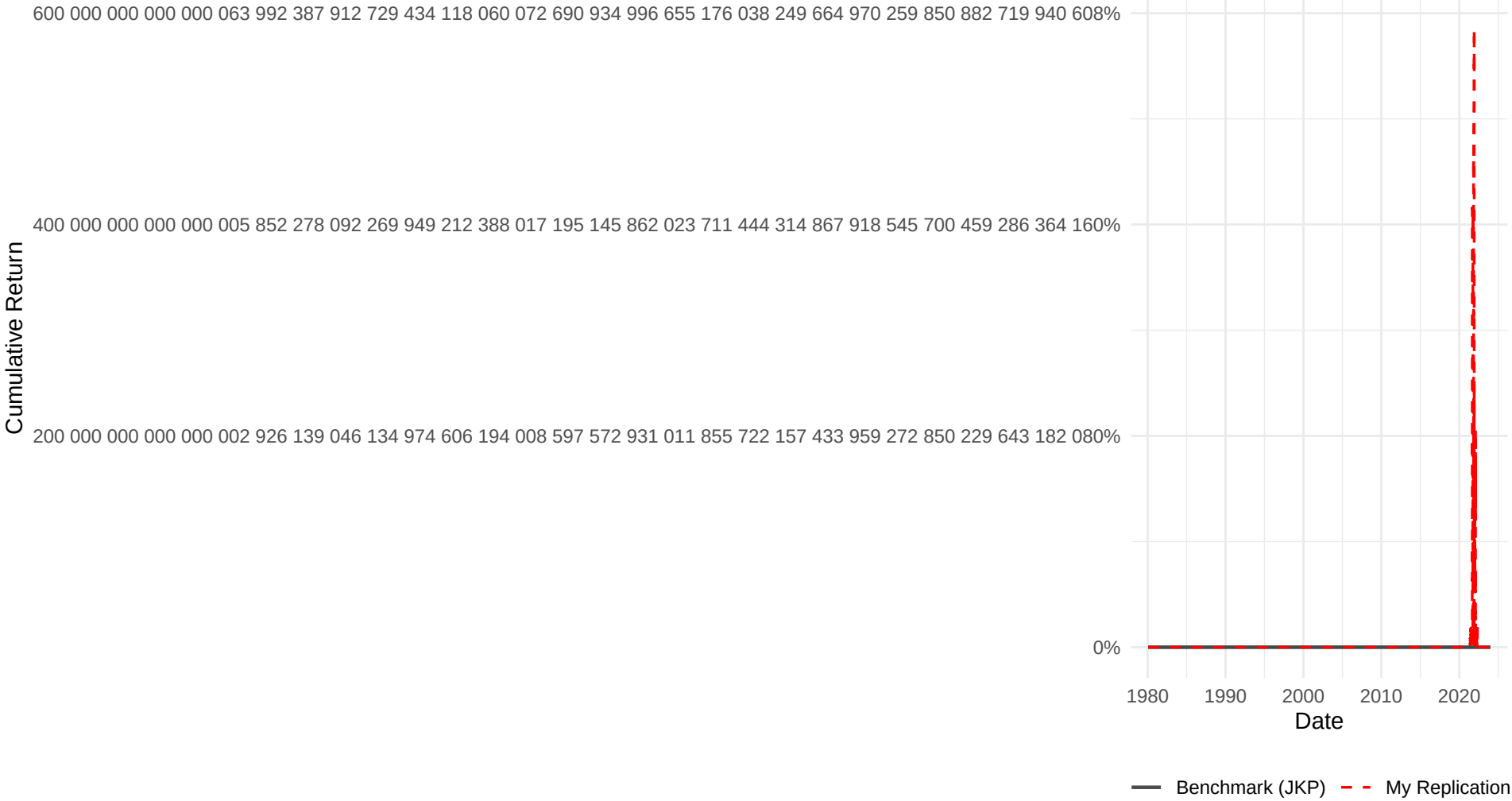
1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: ope_bel1 (Cumulative

Monthly Corr: -0.1184 | Drift Check (P



Factor: debt_gr3 (Cumulative)

Monthly Corr: -0.1142 | Drift Coef: 0.0000

Cumulative Return

39 999 999 999 999 995 945 487 417 118 957 671 753 062 482 563 682 179 808 171 718 291 416 470 542 708 044 041 996 468 224%

30 000 000 000 000 000 577 618 351 505 349 360 801 390 143 444 258 755 270 815 809 519 829 979 140 080 533 278 782 652 416%

19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%

9 999 999 999 999 998 986 371 854 279 739 417 938 265 620 640 920 544 952 042 929 572 854 117 635 677 011 010 499 117 056%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replication



Factor: be_gr1a (Cum

Monthly Corr: -0.1065 | Drift

Cumulative Return

7 500 000 000 000 000 151 641 593 453 669 602 414 320 722 424 107 683 058 533 326 421 560 030 037 486 232 320 190 233 706 496%

5 000 000 000 000 000 101 094 395 635 779 734 942 880 481 616 071 788 705 688 884 281 040 020 024 990 821 546 793 489 137 664%

2 500 000 000 000 000 050 547 197 817 889 867 471 440 240 808 035 894 352 844 442 140 520 010 012 495 410 773 396 744 568 832%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: seas_2

Monthly Corr: -0.104

Cumulative Return

74 999 999 999 999 982 737 362 357 087 567 679 777 817 949 406 159 997 320 494 985 687 783 338 899 059 965 436 510 315 444 445 380 608%

49 999 999 999 999 998 852 475 663 262 266 831 422 342 135 996 207 500 306 499 798 736 599 672 609 039 495 565 163 064 724 075 577 344%

24 999 999 999 999 999 426 237 831 631 133 415 711 171 067 998 103 750 153 249 899 368 299 836 304 519 747 782 581 532 362 037 788 672%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My R

Factor: rskew_21d (Cu
Monthly Corr: -0.103 | Drift Ch

Cumulative Return

299 999 999 999 999 999 955 117 144 473 727 658 110 201 595 493 141 627 866 902 539 803 980 553 024 138 112 329 325 832 306 688%

199 999 999 999 999 999 979 727 437 085 594 788 358 765 312 412 818 410 899 040 858 591 457 082 352 713 540 220 209 982 341 120%

99 999 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replica

Factor: ncoa_gr1a (C

Monthly Corr: -0.1016 | Drift

Cumulative Return

4 000 000 000 000 000 173 509 187 898 476 744 293 161 173 299 807 757 247 167 095 157 344 467 251 558 724 443 765 295 022 080%

3 000 000 000 000 000 245 923 980 161 173 753 643 441 864 983 543 725 788 645 306 033 648 914 478 126 627 340 737 100 906 496%

2 000 000 000 000 000 086 754 593 949 238 372 146 580 586 649 903 878 623 583 547 578 672 233 625 779 362 221 882 647 511 040%

1 000 000 000 000 000 043 377 296 974 619 186 073 290 293 324 951 939 311 791 773 789 336 116 812 889 681 110 941 323 755 520%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: noa_at

Monthly Corr: -0.098

Cumulative Return

15 000 000 000 000 000 001 209 877 812 759 263 306 162 272 166 257 677 375 405 875 410 862 691 518 784 161 776 460 672 347 581 389 471 744%

10 000 000 000 000 000 002 101 697 803 323 328 251 387 822 715 387 464 188 032 188 166 609 887 360 023 982 790 799 717 755 191 065 313 280%

5 000 000 000 000 000 001 050 848 901 661 664 125 693 911 357 693 732 094 016 094 083 304 943 680 011 991 395 399 858 877 595 532 656 640%

0%

19801990200020102020

Date

Benchmark (JKP) My R

Factor: aliq_mat (Cumulative)

Monthly Corr: -0.0979 | Drift

Cumulative Return

4 000 000 000 000 000 173 509 187 898 476 744 293 161 173 299 807 757 247 167 095 157 344 467 251 558 724 443 765 295 022 080%

2 000 000 000 000 000 086 754 593 949 238 372 146 580 586 649 903 878 623 583 547 578 672 233 625 779 362 221 882 647 511 040%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replica

Factor: cowc_gr1a (Cumulative)

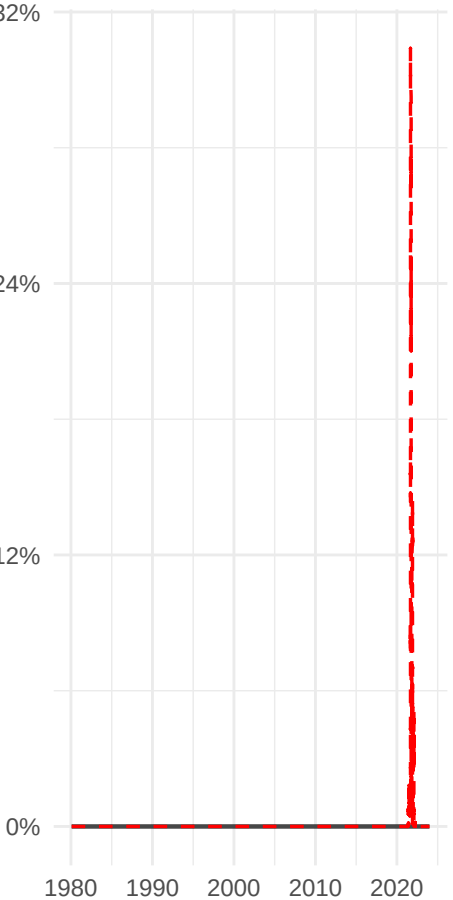
Monthly Corr: -0.0934 | Drift CH

Cumulative Return

60 000 000 000 000 001 155 236 703 010 698 721 602 780 286 888 517 510 541 631 619 039 659 958 280 161 066 557 565 304 832%

39 999 999 999 999 995 945 487 417 118 957 671 753 062 482 563 682 179 808 171 718 291 416 470 542 708 044 041 996 468 224%

19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%



— Benchmark (JKP) - - My Replication

Factor: nfna_gr1a (Cum

Monthly Corr: -0.0908 | Drift Ch

Cumulative Return

1 499 999 999 999 999 893 187 063 000 287 551 528 072 259 115 156 795 747 990 027 195 943 962 973 264 670 404 665 933 824%

999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

499 999 999 999 999 983 242 056 357 731 950 024 912 593 046 310 062 751 489 837 298 654 589 877 718 689 615 343 255 552%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: seas_11_15na (Cumulative Wealth)

Monthly Corr: -0.0865 | Drift Check (Red should overlap Black)

Cumulative Return

19 999 999 999 999 998 987 742 705 941 480 377 339 272 900 220 268 201 461 678 080%

14 999 999 999 999 999 954 430 875 809 090 223 533 597 659 889 948 719 287 631 872%

9 999 999 999 999 999 493 871 352 970 740 188 669 636 450 110 134 100 730 839 040%

4 999 999 999 999 999 746 935 676 485 370 094 334 818 225 055 067 050 365 419 520%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication



Factor: pi_nix (Cu

Monthly Corr: -0.0764

Cumulative Return

300 000 000 000 000 014 484 211 794 056 620 768 648 133 791 035 952 974 905 474 021 993 132 769 424 173 480 518 925 525 601 746 944%

199 999 999 999 999 999 538 074 049 028 741 601 393 225 095 984 807 677 841 113 727 932 195 173 096 259 352 955 823 864 957 370 368%

99 999 999 999 999 999 769 037 024 514 370 800 696 612 547 992 403 838 920 556 863 966 097 586 548 129 676 477 911 932 478 685 184%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Rep

Factor: lti_gr1a (Cumulat

Monthly Corr: -0.0737 | Drift Chec

Cumulative Return

599 999 999 999 999 912 043 540 341 788 381 773 896 487 627 044 004 294 012 423 118 361 739 861 392 749 408 775 307 264%

399 999 999 999 999 997 901 466 300 767 219 729 263 178 441 802 728 702 487 766 778 927 633 234 153 231 380 547 371 008%

199 999 999 999 999 998 950 733 150 383 609 864 631 589 220 901 364 351 243 883 389 463 816 617 076 615 690 273 685 504%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

100 000 000 000 000 001 440 594 758 724 527 385 583 111 862 242 831 263 013 712 314 935 498 927 069 126 131 626 863 257 625 726 456 080 505 437 183 296 233 537 5369

50 000 000 000 000 000 720 297 379 362 263 692 791 555 931 121 415 631 506 856 157 467 749 463 534 563 065 813 431 628 812 863 228 040 252 718 591 648 116 768 7689

09

Factor: dolvol_var_1

Monthly Corr: -0.0715 | Drift

Cumulative Return

40 000 000 000 000 000 808 755 165 086 237 879 543 043 852 928 574 309 645 511 074 248 320 160 199 926 572 374 347 913 101 312%

30 000 000 000 000 000 606 566 373 814 678 409 657 282 889 696 430 732 234 133 305 686 240 120 149 944 929 280 760 934 825 984%

20 000 000 000 000 000 404 377 582 543 118 939 771 521 926 464 287 154 822 755 537 124 160 080 099 963 286 187 173 956 550 656%

10 000 000 000 000 000 202 188 791 271 559 469 885 760 963 232 143 577 411 377 768 562 080 040 049 981 643 093 586 978 275 328%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: iskew_capm

Monthly Corr: -0.0714 | Drift

Cumulative Return

2 499 999 999 999 999 587 378 840 868 625 085 777 156 300 773 284 262 939 764 503 477 957 753 854 665 074 741 744 226 009 088%

1 999 999 999 999 999 855 170 415 474 605 981 299 438 616 632 528 062 917 043 578 247 391 105 546 864 194 206 056 388 231 168%

1 499 999 999 999 999 891 377 811 605 954 485 974 578 962 474 396 047 187 782 683 685 543 329 160 148 145 654 542 291 173 376%

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: qmj_growth

Monthly Corr: -0.0695 | Dr

Cumulative Return

149 999 999 999 999 995 622 138 157 885 155 541 177 871 407 926 127 558 561 387 509 830 204 502 224 439 269 897 364 377 174 016%

100 000 000 000 000 002 021 887 912 715 594 698 857 609 632 321 435 774 113 777 685 620 800 400 499 816 430 935 869 782 753 280%

50 000 000 000 000 001 010 943 956 357 797 349 428 804 816 160 717 887 056 888 842 810 400 200 249 908 215 467 934 891 376 640%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replic

Factor: fnl_gr1a (Cumul

Monthly Corr: -0.0659 | Drift Ch

Cumulative Return

5 999 999 999 999 999 572 748 252 001 150 206 112 289 036 460 627 182 991 960 108 783 775 851 893 058 681 618 663 735 296%

3 999 999 999 999 999 865 936 450 861 855 600 199 300 744 370 480 502 011 918 698 389 236 719 021 749 516 922 746 044 416%

1 999 999 999 999 999 932 968 225 430 927 800 099 650 372 185 240 251 005 959 349 194 618 359 510 874 758 461 373 022 208%

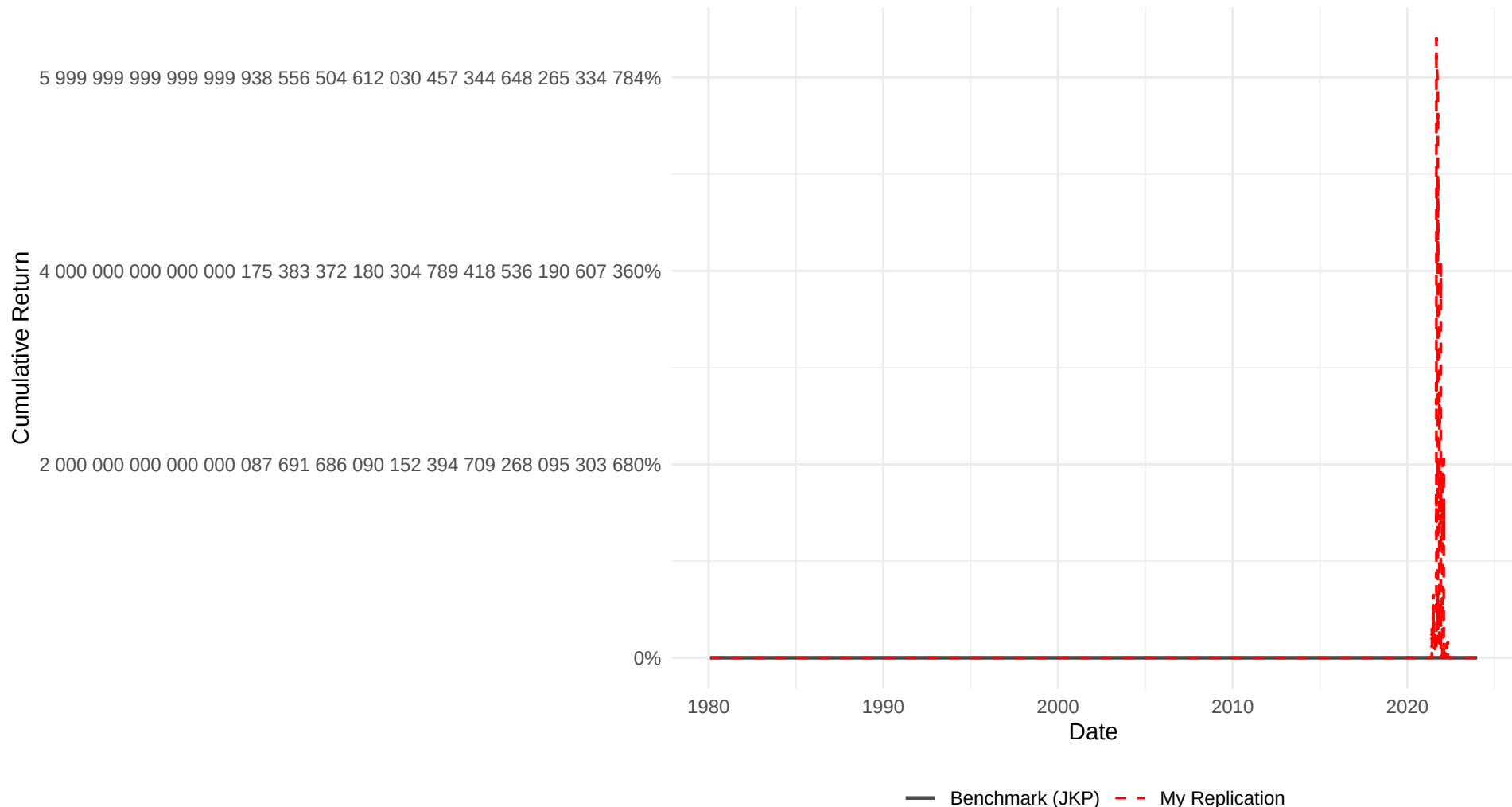
0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: saleq_su (Cumulative Wealth)

Monthly Corr: -0.0629 | Drift Check (Red should overlap Black)



Factor: dsale_drec (C

Monthly Corr: -0.0626 | Drift

Cumulative Return

5 000 000 000 000 000 101 094 395 635 779 734 942 880 481 616 071 788 705 688 884 281 040 020 024 990 821 546 793 489 137 664%

4 000 000 000 000 000 173 509 187 898 476 744 293 161 173 299 807 757 247 167 095 157 344 467 251 558 724 443 765 295 022 080%

3 000 000 000 000 000 245 923 980 161 173 753 643 441 864 983 543 725 788 645 306 033 648 914 478 126 627 340 737 100 906 496%

2 000 000 000 000 000 086 754 593 949 238 372 146 580 586 649 903 878 623 583 547 578 672 233 625 779 362 221 882 647 511 040%

1 000 000 000 000 000 043 377 296 974 619 186 073 290 293 324 951 939 311 791 773 789 336 116 812 889 681 110 941 323 755 520%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replica

Factor: iskew_ff3_21d (Cumulative Wealth)
Monthly Corr: -0.0591 | Drift Check (Red should overlap Black)

Cumulative Return

39 999 999 999 999 997 975 485 411 882 960 754 678 545 800 440 536 402 923 356 160%
29 999 999 999 999 999 908 861 751 618 180 447 067 195 319 779 897 438 575 263 744%
19 999 999 999 999 998 987 742 705 941 480 377 339 272 900 220 268 201 461 678 080%
9 999 999 999 999 999 493 871 352 970 740 188 669 636 450 110 134 100 730 839 040%
0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: gp_at (Cumulative W

Monthly Corr: -0.0556 | Drift Check (R

Cumulative Return

60 000 000 000 000 000 877 841 713 840 492 381 858 202 579 271 879 303 556 716 647 230 187 781 855 068 892 954 624%

40 000 000 000 000 000 585 227 809 226 994 921 238 801 719 514 586 202 371 144 431 486 791 854 570 045 928 636 416%

20 000 000 000 000 000 292 613 904 613 497 460 619 400 859 757 293 101 185 572 215 743 395 927 285 022 964 318 208%

0%

1980 1990 2000 2010 2020
Date

Benchmark (JKP) My Replication

Factor: dsale_dinv (Cum

Monthly Corr: -0.052 | Drift Chec

Cumulative Return

2 999 999 999 999 999 999 786 374 126 000 575 103 056 144 518 230 313 591 495 980 054 391 887 925 946 529 340 809 331 867 648%

1 999 999 999 999 999 999 932 968 225 430 927 800 099 650 372 185 240 251 005 959 349 194 618 359 510 874 758 461 373 022 208%

999 999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: op_atl1 (Cumula

Monthly Corr: -0.0483 | Drift Che

Cumulative Return

7 500 000 000 000 000 144 404 587 876 337 340 200 347 535 861 064 688 817 703 952 379 957 494 785 020 133 319 695 663 104%

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

2 499 999 999 999 999 746 592 963 569 934 854 484 566 405 160 230 136 238 010 732 393 213 529 408 919 252 752 624 779 264%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati



Factor: ebitda_m

Monthly Corr: -0.0344

Cumulative Return

399 999 999 999 999 999 076 148 098 057 483 202 786 450 191 969 615 355 682 227 455 864 390 346 192 518 705 911 647 729 914 740 736%

300 000 000 000 000 000 014 484 211 794 056 620 768 648 133 791 035 952 974 905 474 021 993 132 769 424 173 480 518 925 525 601 746 944%

199 999 999 999 999 999 538 074 049 028 741 601 393 225 095 984 807 677 841 113 727 932 195 173 096 259 352 955 823 864 957 370 368%

99 999 999 999 999 999 769 037 024 514 370 800 696 612 547 992 403 838 920 556 863 966 097 586 548 129 676 477 911 932 478 685 184%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Rep



Factor: dolvol_12

Monthly Corr: -0.029 |

Cumulative Return

199 999 999 999 999 999 538 074 049 028 741 601 393 225 095 984 807 677 841 113 727 932 195 173 096 259 352 955 823 864 957 370 368%

99 999 999 999 999 999 769 037 024 514 370 800 696 612 547 992 403 838 920 556 863 966 097 586 548 129 676 477 911 932 478 685 184%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Rep

Factor: seas_11_15an (Cumulative Wealth)

Monthly Corr: -0.0272 | Drift Check (Red should overlap Black)

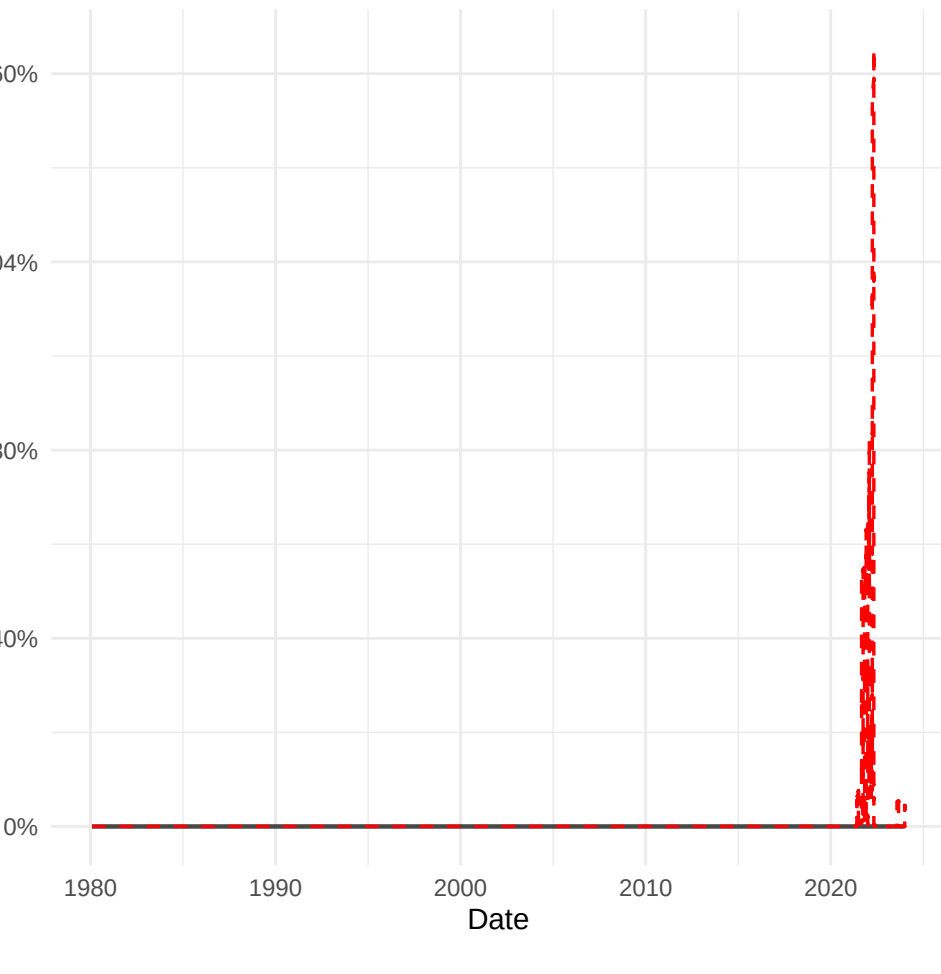
Cumulative Return

400 000 000 000 000 033 276 642 595 303 103 086 471 181 858 761 431 643 586 560%
300 000 000 000 000 002 656 736 747 946 704 173 317 668 121 422 712 226 709 504%
200 000 000 000 000 016 638 321 297 651 551 543 235 590 929 380 715 821 793 280%
100 000 000 000 000 008 319 160 648 825 775 771 617 795 464 690 357 910 896 640%
0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication



Factor: **dgp_dsale** (Cumulative

Monthly Corr: **-0.0215** | Drift Check (Red

Cumulative Return

149 999 999 999 999 970 921 066 150 706 488 792 831 768 947 236 926 677 472 704 087 928 405 376 557 484 343 296%

99 999 999 999 999 989 600 692 989 521 205 793 443 517 660 497 828 009 527 517 532 799 127 744 739 526 311 936%

49 999 999 999 999 994 800 346 494 760 602 896 721 758 830 248 914 004 763 758 766 399 563 872 369 763 155 968%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication

Factor: coskew_21d (C
Monthly Corr: -0.0195 | Drift C

Cumulative Return

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

749 999 999 999 999 945 688 905 802 977 242 987 289 481 237 198 023 593 891 341 842 771 664 580 074 072 827 271 145 586 688%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

249 999 999 999 999 981 896 301 934 325 747 662 429 827 079 066 007 864 630 447 280 923 888 193 358 024 275 757 048 528 896%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replicat

Factor: eq_dur (Cumula

Monthly Corr: -0.0191 | Drift CH

Cumulative Return

24 999 999 999 999 997 465 929 635 699 348 544 845 664 051 602 301 362 380 107 323 932 135 294 089 192 527 526 247 792 640%

19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%

15 000 000 000 000 000 000 288 809 175 752 674 680 400 695 071 722 129 377 635 407 904 759 914 989 570 040 266 639 391 326 208%

9 999 999 999 999 998 986 371 854 279 739 417 938 265 620 640 920 544 952 042 929 572 854 117 635 677 011 010 499 117 056%

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: capex_abn (Cumulative

Monthly Corr: -0.0157 | Drift Check (Re

Cumulative Return

7 500 000 000 000 000 109 730 214 230 061 547 732 275 322 408 984 912 944 589 580 903 773 472 731 883 611 619 328%

4 999 999 999 999 999 210 435 182 804 553 891 725 505 732 154 695 093 740 004 432 414 550 662 425 940 213 104 640%

2 499 999 999 999 999 605 217 591 402 276 945 862 752 866 077 347 546 870 002 216 207 275 331 212 970 106 552 320%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: rmax5_rvol_21d (Cu

Monthly Corr: -0.0129 | Drift Check (P

Cumulative Return

999 999 999 999 999 959 416 724 456 350 362 731 491 996 089 648 451 439 669 739 009 806 703 922 950 954 425 516 032%

499 999 999 999 999 979 708 362 228 175 181 365 745 998 044 824 225 719 834 869 504 903 351 961 475 477 212 758 016%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: iskew_hxz4_21d (Cumulative Wealth)

Monthly Corr: -0.0081 | Drift Check (Red should overlap Black)

Cumulative Return

20 000 000 000 000 002 398 097 581 175 399 228 888 724 799 488%

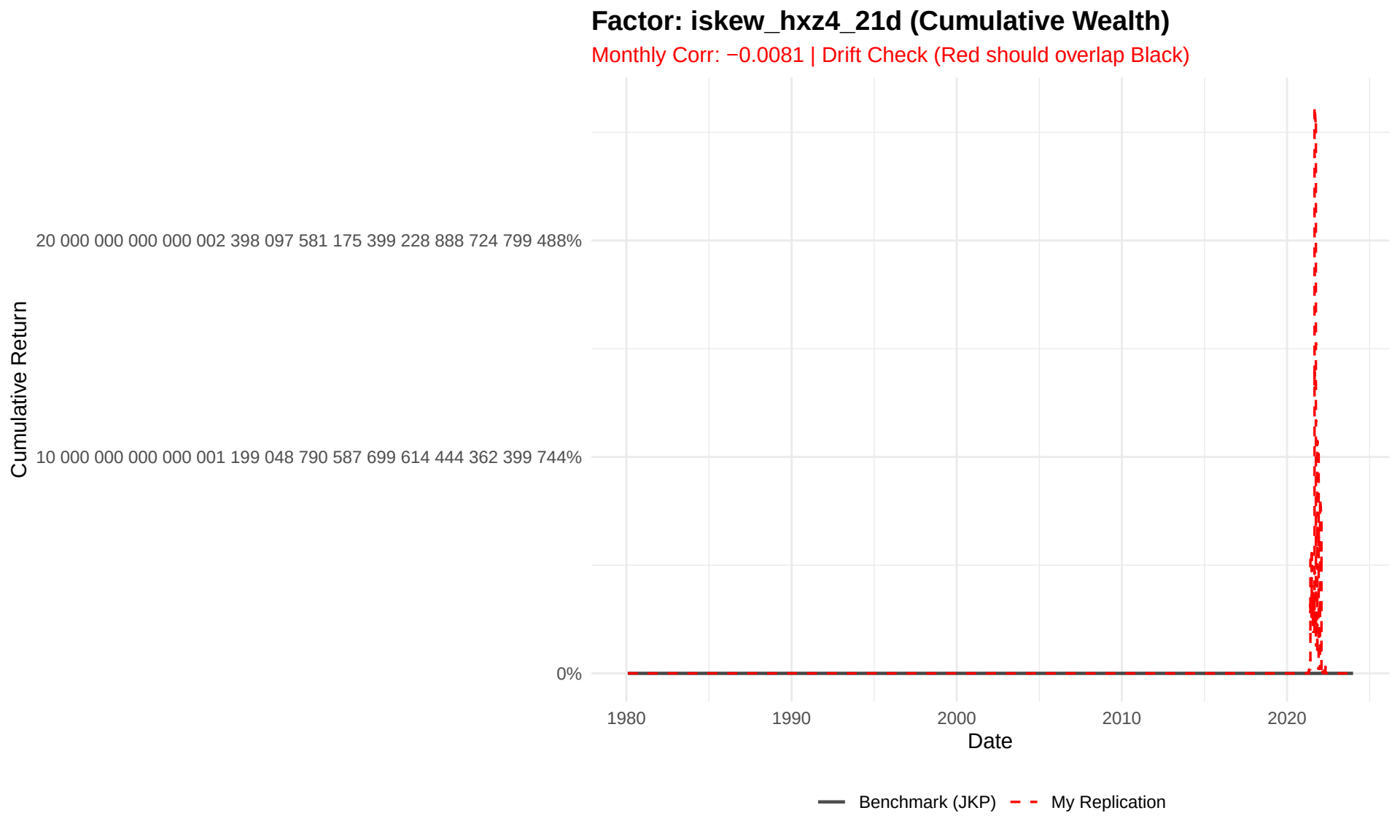
10 000 000 000 000 001 199 048 790 587 699 614 444 362 399 744%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication



Factor: debt_me (Cumulative)

Monthly Corr: -0.0035 | Drift Check

Cumulative Return

5 999 999 999 999 999 756 500 346 738 102 176 388 951 976 537 890 708 638 018 434 058 840 223 537 705 726 553 096 192%

3 999 999 999 999 999 837 666 897 825 401 450 925 967 984 358 593 805 758 678 956 039 226 815 691 803 817 702 064 128%

1 999 999 999 999 999 918 833 448 912 700 725 462 983 992 179 296 902 879 339 478 019 613 407 845 901 908 851 032 064%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: z_score (Cumulative)

Monthly Corr: 0.0019 | Drift Check

Cumulative Return

1 999 999 999 999 999 932 968 225 430 927 800 099 650 372 185 240 251 005 959 349 194 618 359 510 874 758 461 373 022 208%

1 499 999 999 999 999 893 187 063 000 287 551 528 072 259 115 156 795 747 990 027 195 943 962 973 264 670 404 665 933 824%

999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

499 999 999 999 999 983 242 056 357 731 950 024 912 593 046 310 062 751 489 837 298 654 589 877 718 689 615 343 255 552%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: at_turnover (

Monthly Corr: 0.0052 | Drift

Cumulative Return

20 000 000 000 000 000 404 377 582 543 118 939 771 521 926 464 287 154 822 755 537 124 160 080 099 963 286 187 173 956 550 656%

10 000 000 000 000 000 202 188 791 271 559 469 885 760 963 232 143 577 411 377 768 562 080 040 049 981 643 093 586 978 275 328%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: at_me (Cumulative

Monthly Corr: 0.0144 | Drift Check

Cumulative Return

900 000 000 000 000 094 221 734 804 315 766 847 506 811 535 659 576 466 936 573 477 621 836 431 654 717 878 618 292 224%

599 999 999 999 999 912 043 540 341 788 381 773 896 487 627 044 004 294 012 423 118 361 739 861 392 749 408 775 307 264%

299 999 999 999 999 956 021 770 170 894 190 886 948 243 813 522 002 147 006 211 559 180 869 930 696 374 704 387 653 632%

0%

Date

— Benchmark (JKP) - - My Replication

Factor: seas_1_1

Monthly Corr: 0.0149 |

Cumulative Return

300 000 000 000 000 014 484 211 794 056 620 768 648 133 791 035 952 974 905 474 021 993 132 769 424 173 480 518 925 525 601 746 944%

199 999 999 999 999 999 538 074 049 028 741 601 393 225 095 984 807 677 841 113 727 932 195 173 096 259 352 955 823 864 957 370 368%

99 999 999 999 999 999 769 037 024 514 370 800 696 612 547 992 403 838 920 556 863 966 097 586 548 129 676 477 911 932 478 685 184%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Rep

Factor: gp_atl1 (Cumulative

Monthly Corr: 0.0178 | Drift Check (f

Cumulative Return

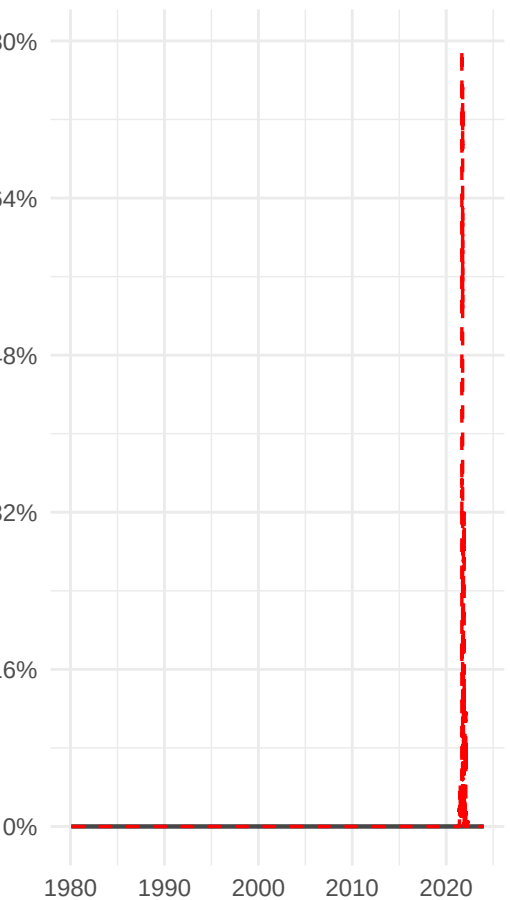
2 499 999 999 999 999 898 541 811 140 875 906 828 729 990 224 121 128 599 174 347 524 516 759 807 377 386 063 790 080%

1 999 999 999 999 999 918 833 448 912 700 725 462 983 992 179 296 902 879 339 478 019 613 407 845 901 908 851 032 064%

1 499 999 999 999 999 939 125 086 684 525 544 097 237 994 134 472 677 159 504 608 514 710 055 884 426 431 638 274 048%

999 999 999 999 999 959 416 724 456 350 362 731 491 996 089 648 451 439 669 739 009 806 703 922 950 954 425 516 032%

499 999 999 999 999 979 708 362 228 175 181 365 745 998 044 824 225 719 834 869 504 903 351 961 475 477 212 758 016%



— Benchmark (JKP) - - My Replication

Factor: ocf_at_chg1 (Cu

Monthly Corr: 0.0193 | Drift Chec

Cumulative Return

7 500 000 000 000 000 144 404 587 876 337 340 200 347 535 861 064 688 817 703 952 379 957 494 785 020 133 319 695 663 104%

4 999 999 999 999 999 493 185 927 139 869 708 969 132 810 320 460 272 476 021 464 786 427 058 817 838 505 505 249 558 528%

2 499 999 999 999 999 746 592 963 569 934 854 484 566 405 160 230 136 238 010 732 393 213 529 408 919 252 752 624 779 264%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replicati

Factor: tax_gr1a (Cum

Monthly Corr: 0.0225 | Drift Ch

Cumulative Return

149 999 999 999 999 977 558 572 236 863 829 055 100 797 746 570 813 933 451 269 901 990 276 512 069 056 164 662 916 153 344%

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicat

Factor: seas_6_10an (Cumulative)

Monthly Corr: 0.0237 | Drift Check (Red sho

Cumulative Return

2 499 999 999 999 999 908 516 991 407 721 643 552 756 785 806 318 391 948 420 090 960 856 771 625 385 721 856%

1 999 999 999 999 999 842 563 759 791 331 565 483 871 006 498 118 367 703 619 996 448 246 128 296 859 795 456%

1 499 999 999 999 999 776 610 528 174 941 487 414 985 227 189 918 343 458 819 901 935 635 484 968 333 869 056%

999 999 999 999 999 921 281 879 895 665 782 741 935 503 249 059 183 851 809 998 224 123 064 148 429 897 728%

499 999 999 999 999 960 640 939 947 832 891 370 967 751 624 529 591 925 904 999 112 061 532 074 214 948 864%

0%

1980 1990 2000 2010 2020

Date

Benchmark (JKP) My Replication

Factor: cop_atl1 (Cumula

Monthly Corr: 0.028 | Drift Check (

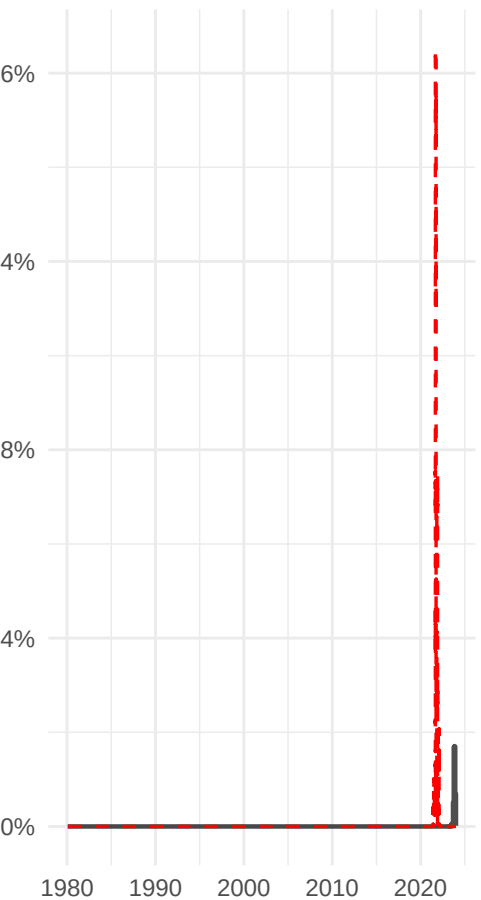
Cumulative Return

799 999 999 999 999 995 802 932 601 534 439 458 526 356 883 605 457 404 975 533 557 855 266 468 306 462 761 094 742 016%

599 999 999 999 999 999 912 043 540 341 788 381 773 896 487 627 044 004 294 012 423 118 361 739 861 392 749 408 775 307 264%

399 999 999 999 999 999 997 901 466 300 767 219 729 263 178 441 802 728 702 487 766 778 927 633 234 153 231 380 547 371 008%

199 999 999 999 999 999 998 950 733 150 383 609 864 631 589 220 901 364 351 243 883 389 463 816 617 076 615 690 273 685 504%

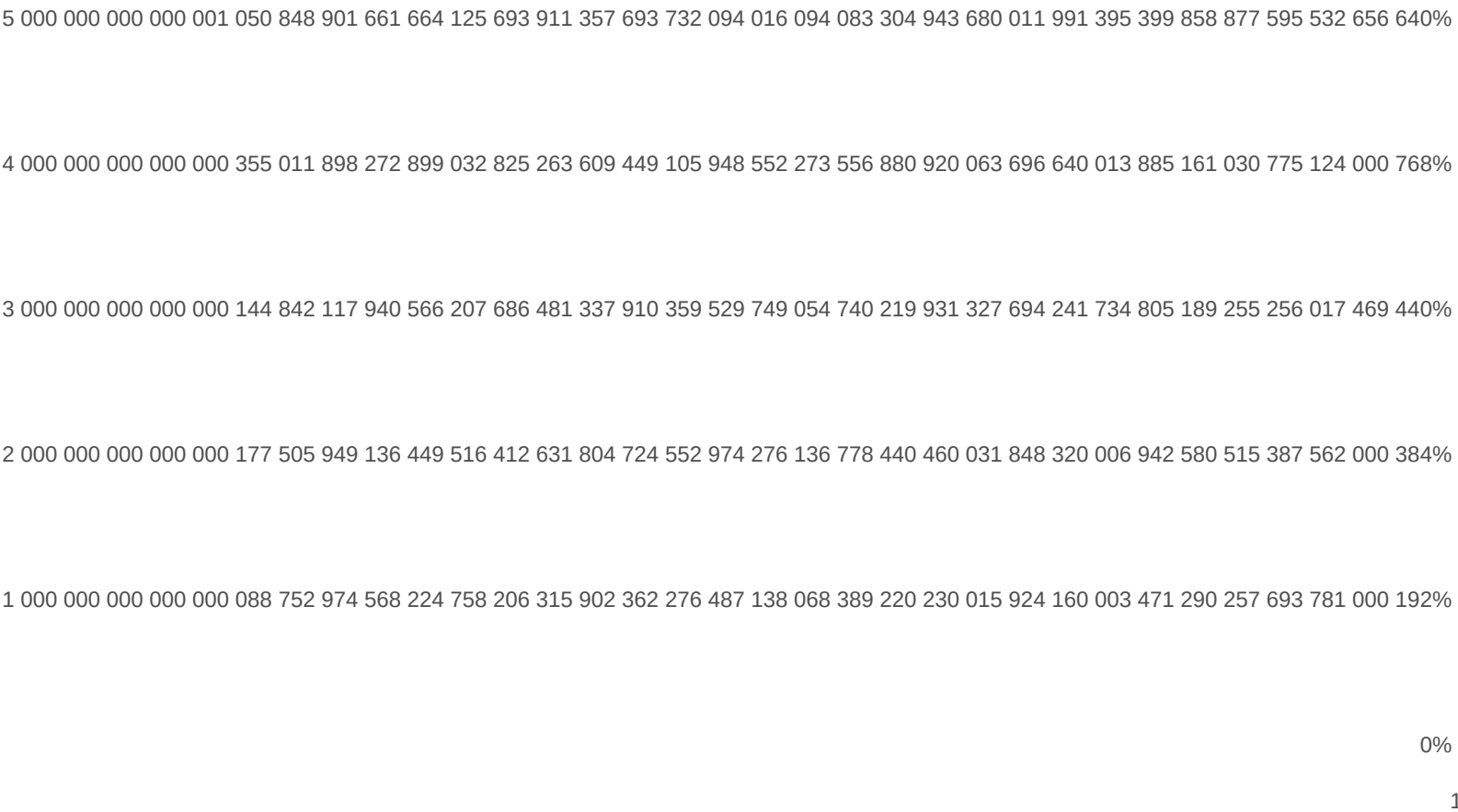


— Benchmark (JKP) - - My Replication

Factor: sale_bev

Monthly Corr: 0.0317

Cumulative Return



— Benchmark (JKP) - - My Re

Factor: bev_mev (Cumulative)

Monthly Corr: 0.0376 | Drift Check: 0.0000

Cumulative Return

2 999 999 999 999 999 999 786 374 126 000 575 103 056 144 518 230 313 591 495 980 054 391 887 925 946 529 340 809 331 867 648%

1 999 999 999 999 999 999 932 968 225 430 927 800 099 650 372 185 240 251 005 959 349 194 618 359 510 874 758 461 373 022 208%

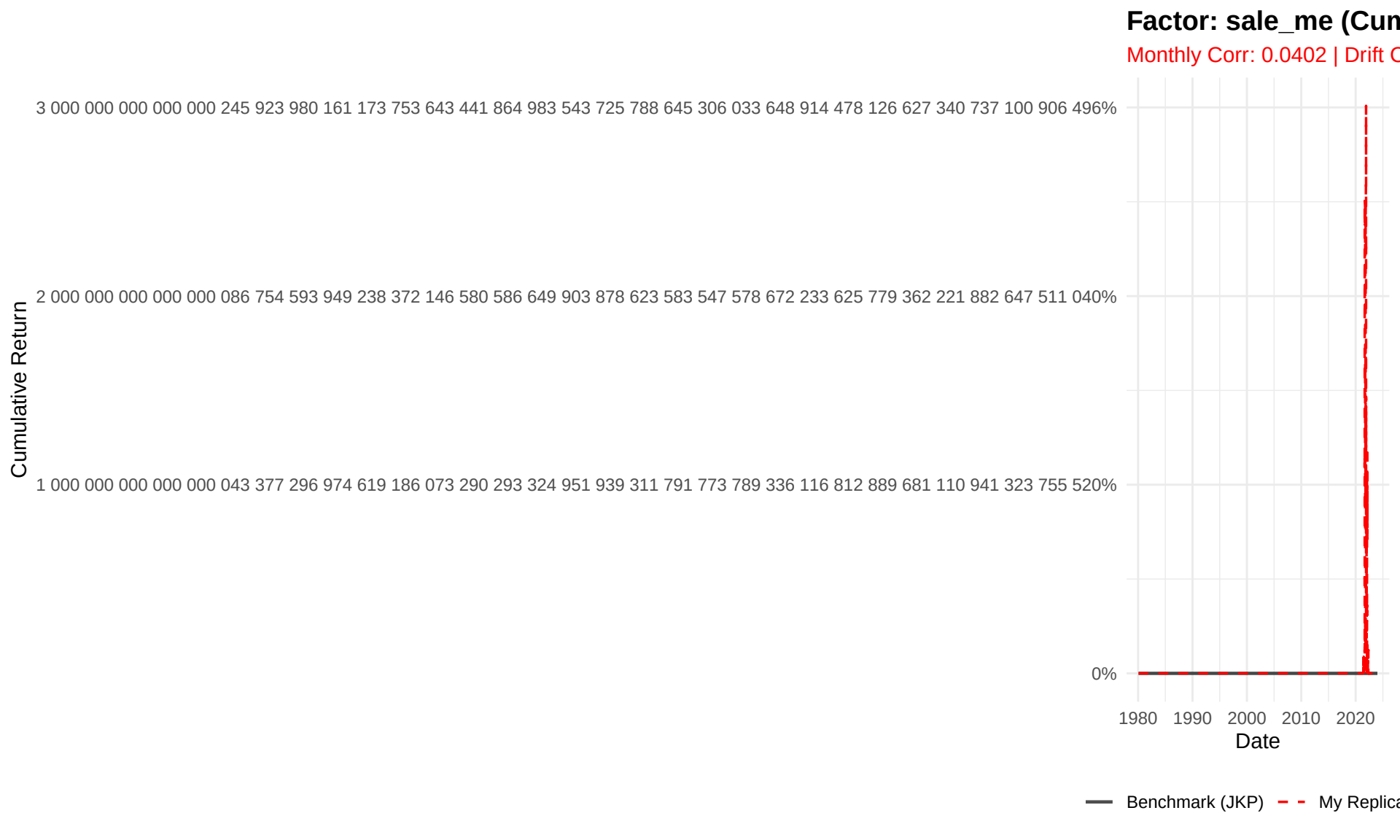
999 999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

0%

1980 1990 2000 2010 2020

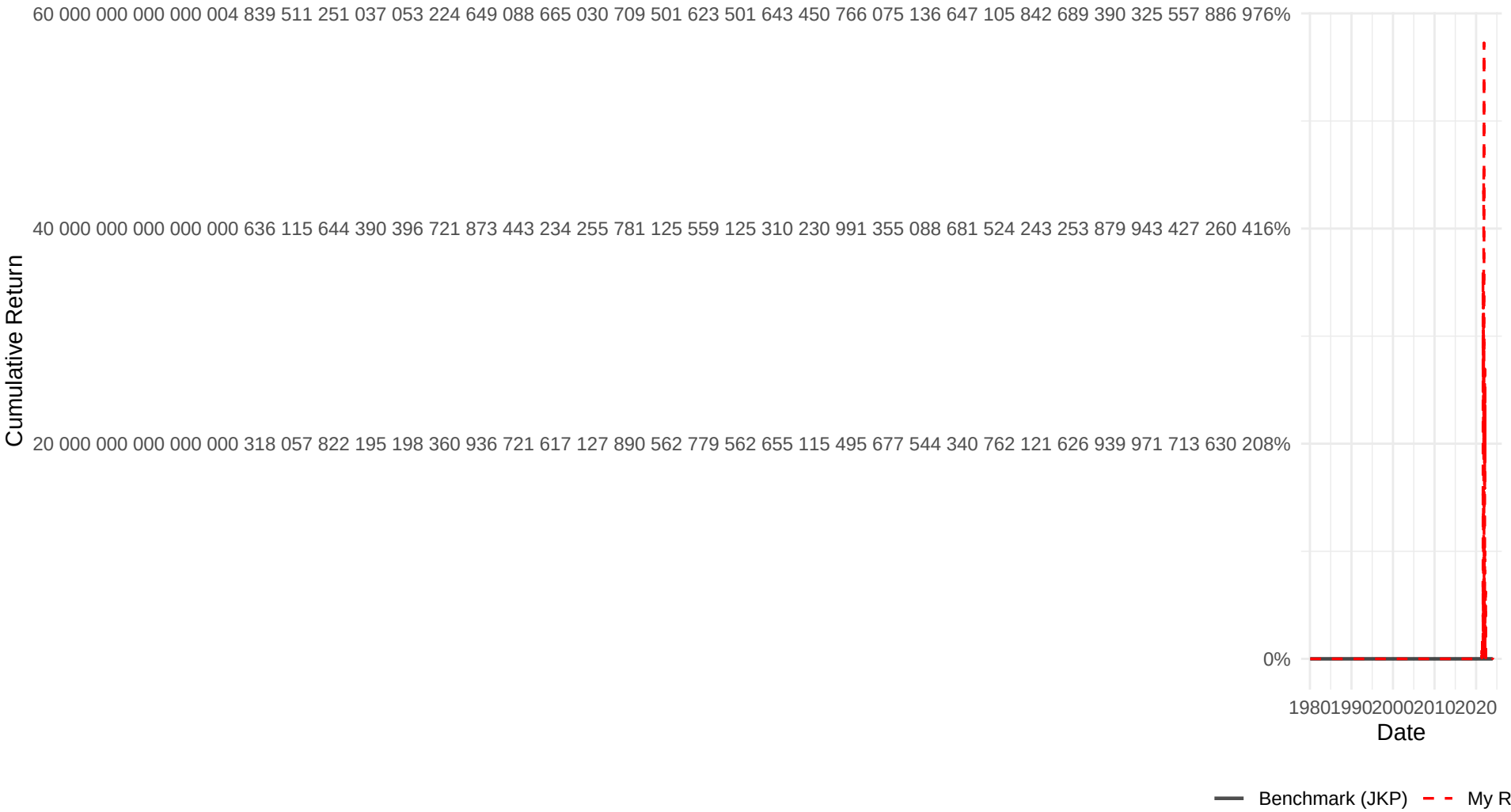
Date

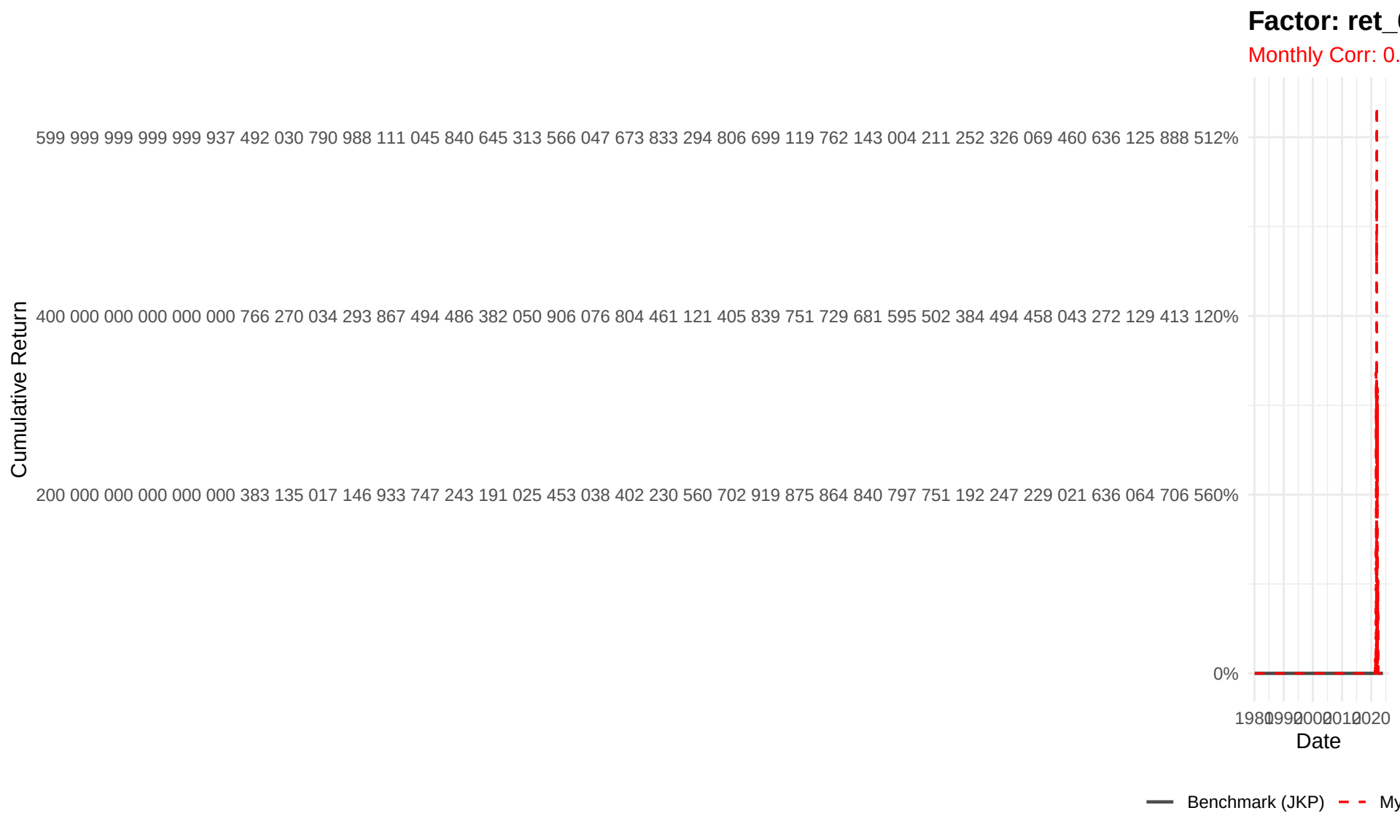
— Benchmark (JKP) - - My Replication



Factor: seas_2

Monthly Corr: 0.0408





Factor: be_me (Cum

Monthly Corr: 0.0449 | Drift

Cumulative Return

200 000 000 000 000 004 043 775 825 431 189 397 715 219 264 642 871 548 227 555 371 241 600 800 999 632 861 871 739 565 506 560%

149 999 999 999 999 995 622 138 157 885 155 541 177 871 407 926 127 558 561 387 509 830 204 502 224 439 269 897 364 377 174 016%

100 000 000 000 000 002 021 887 912 715 594 698 857 609 632 321 435 774 113 777 685 620 800 400 499 816 430 935 869 782 753 280%

50 000 000 000 000 001 010 943 956 357 797 349 428 804 816 160 717 887 056 888 842 810 400 200 249 908 215 467 934 891 376 640%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replic

Factor: netdebt_me (Cumulative)

Monthly Corr: 0.047 | Drift Check (P

Cumulative Return

79 999 999 999 999 996 753 337 956 508 029 018 519 359 687 171 876 115 173 579 120 784 536 313 836 076 354 041 282 560%

60 000 000 000 000 008 166 085 856 051 327 741 389 304 769 836 418 181 345 087 721 842 115 984 106 694 473 286 680 576%

39 999 999 999 999 998 376 668 978 254 014 509 259 679 843 585 938 057 586 789 560 392 268 156 918 038 177 020 641 280%

19 999 999 999 999 999 188 334 489 127 007 254 629 839 921 792 969 028 793 394 780 196 134 078 459 019 088 510 320 640%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: opex_at (Cumulat

Monthly Corr: 0.0646 | Drift Check

Cumulative Return

999 999 999 999 999 999 966 484 112 715 463 900 049 825 186 092 620 125 502 979 674 597 309 179 755 437 379 230 686 511 104%

749 999 999 999 999 999 946 593 531 500 143 775 764 036 129 557 578 397 873 995 013 597 971 981 486 632 335 202 332 966 912%

499 999 999 999 999 999 983 242 056 357 731 950 024 912 593 046 310 062 751 489 837 298 654 589 877 718 689 615 343 255 552%

249 999 999 999 999 999 991 621 028 178 865 975 012 456 296 523 155 031 375 744 918 649 327 294 938 859 344 807 671 627 776%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replication

Factor: ni_ar1 (Cumulative

Monthly Corr: 0.0699 | Drift Check

Cumulative Return

60 000 000 000 000 008 166 085 856 051 327 741 389 304 769 836 418 181 345 087 721 842 115 984 106 694 473 286 680 576%

39 999 999 999 999 998 376 668 978 254 014 509 259 679 843 585 938 057 586 789 560 392 268 156 918 038 177 020 641 280%

19 999 999 999 999 999 188 334 489 127 007 254 629 839 921 792 969 028 793 394 780 196 134 078 459 019 088 510 320 640%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: dsale_dsga (Cumulative Wealth)

Monthly Corr: 0.0708 | Drift Check (Red should overlap)

Cumulative Return

200 000 000 000 000 008 376 305 112 842 291 591 798 286 773 328 067 656 628 685 542 361 399 296%

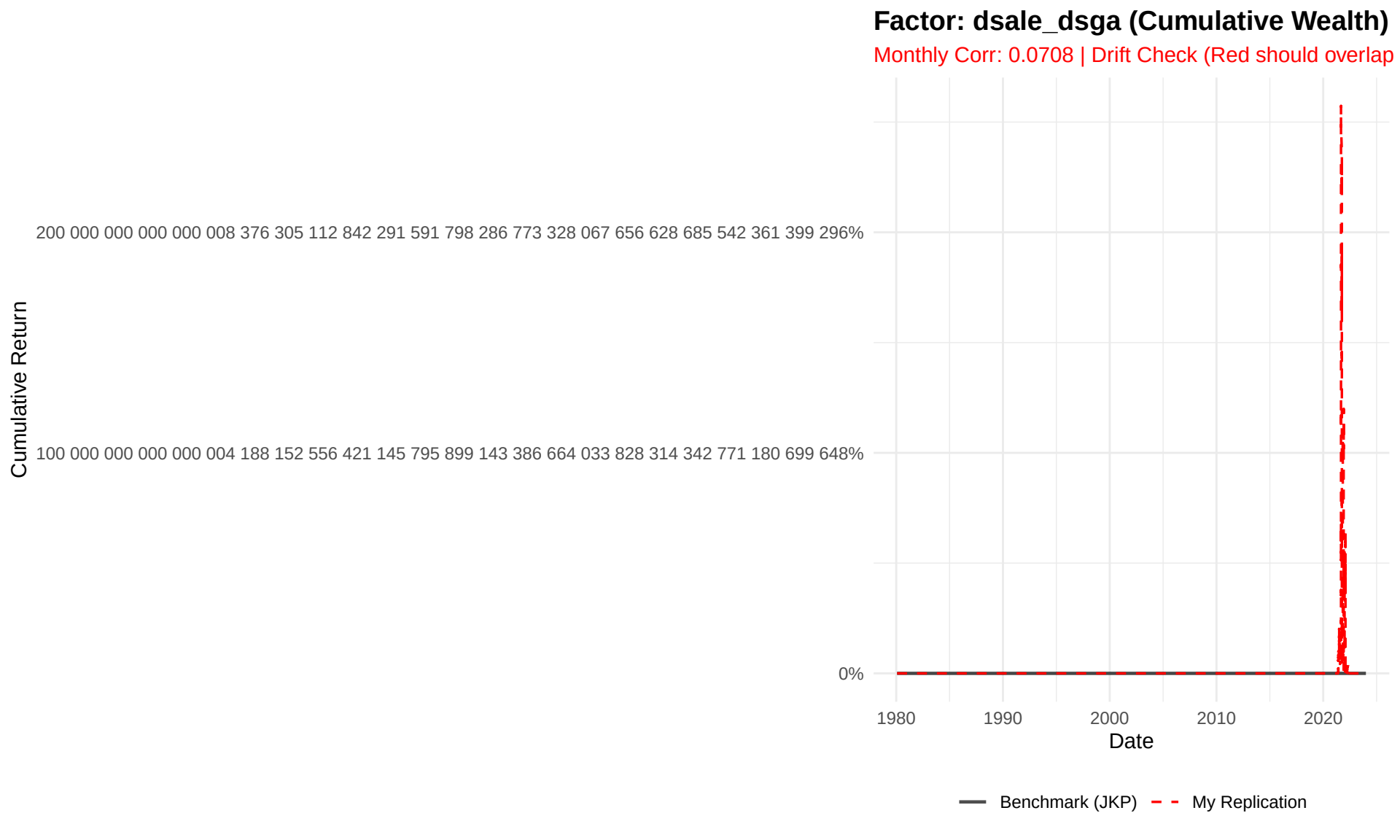
100 000 000 000 000 004 188 152 556 421 145 795 899 143 386 664 033 828 314 342 771 180 699 648%

0%

1980 1990 2000 2010 2020

Date

Benchmark (JKP) My Replication



Factor: sti_gr1a (Cumulative W

Monthly Corr: 0.0725 | Drift Check (Red sh

Cumulative Return

24 999 999 999 999 997 400 173 247 380 301 448 360 879 415 124 457 002 381 879 383 199 781 936 184 881 577 984%

19 999 999 999 999 999 268 135 931 261 773 148 422 054 286 450 547 135 587 360 727 686 854 173 003 085 774 848%

14 999 999 999 999 999 451 101 948 446 329 861 316 540 714 837 910 351 690 520 545 765 140 629 752 314 331 136%

9 999 999 999 999 999 634 067 965 630 886 574 211 027 143 225 273 567 793 680 363 843 427 086 501 542 887 424%

4 999 999 999 999 999 817 033 982 815 443 287 105 513 571 612 636 783 896 840 181 921 713 543 250 771 443 712%

0%

1980 1990 2000 2010 2020

Date

Benchmark (JKP) My Replication

Factor: dbnetis_at (Cur
Monthly Corr: 0.0822 | Drift Che

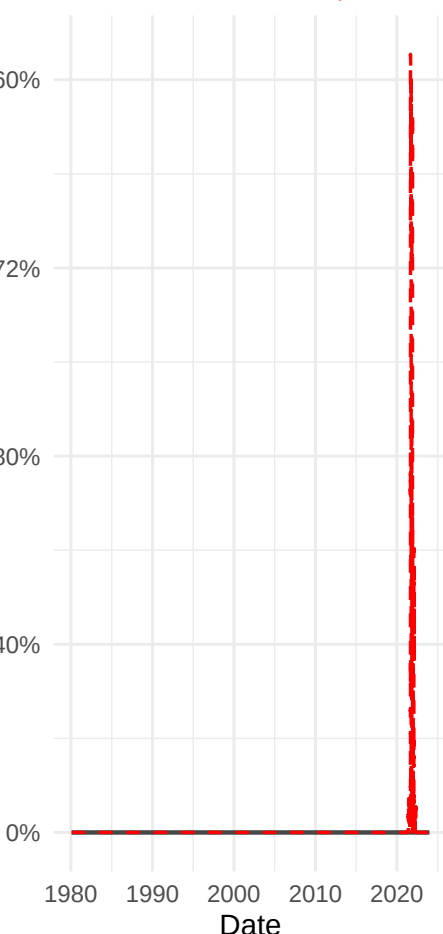
Cumulative Return

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

74 999 999 999 999 988 779 286 118 431 914 527 550 398 873 285 406 966 725 634 950 995 138 256 034 528 082 331 458 076 672%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

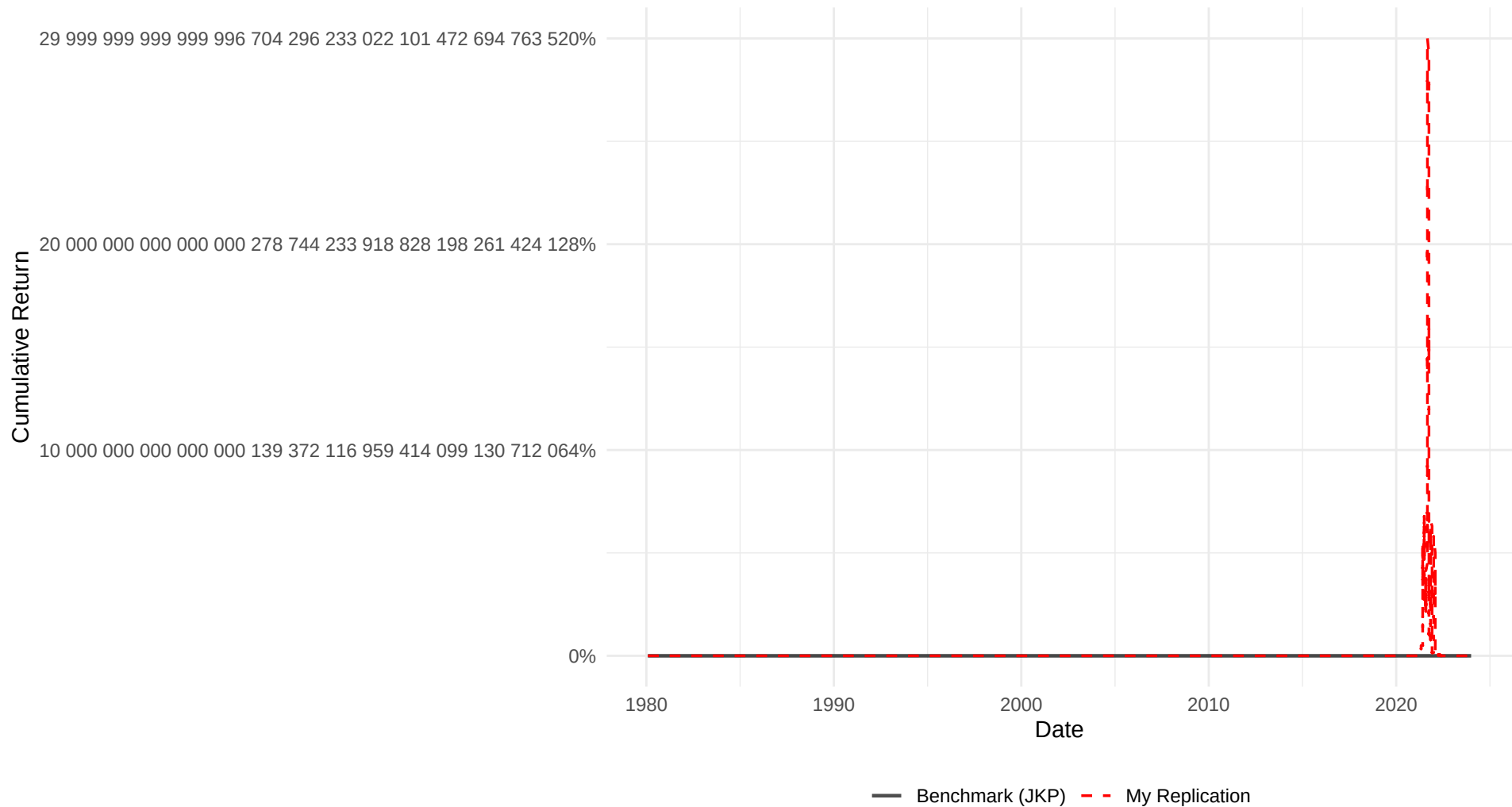
24 999 999 999 999 997 465 929 635 699 348 544 845 664 051 602 301 362 380 107 323 932 135 294 089 192 527 526 247 792 640%



— Benchmark (JKP) - - My Replication

Factor: rd5_at (Cumulative Wealth)

Monthly Corr: 0.0842 | Drift Check (Red should overlap Black)



Factor: ival_me (Cumulative)

Monthly Corr: 0.0908 | Drift Coefficient: 0.0000

Cumulative Return

4 000 000 000 000 000 173 509 187 898 476 744 293 161 173 299 807 757 247 167 095 157 344 467 251 558 724 443 765 295 022 080%

2 000 000 000 000 000 086 754 593 949 238 372 146 580 586 649 903 878 623 583 547 578 672 233 625 779 362 221 882 647 511 040%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: at_be (Cumulative)

Monthly Corr: 0.0917 | Drift Coef: 0.0000

Cumulative Return

1 499 999 999 999 999 891 377 811 605 954 485 974 578 962 474 396 047 187 782 683 685 543 329 160 148 145 654 542 291 173 376%

999 999 999 999 999 927 585 207 737 302 990 649 719 308 316 264 031 458 521 789 123 695 552 773 432 097 103 028 194 115 584%

499 999 999 999 999 963 792 603 868 651 495 324 859 654 158 132 015 729 260 894 561 847 776 386 716 048 551 514 097 057 792%

0%

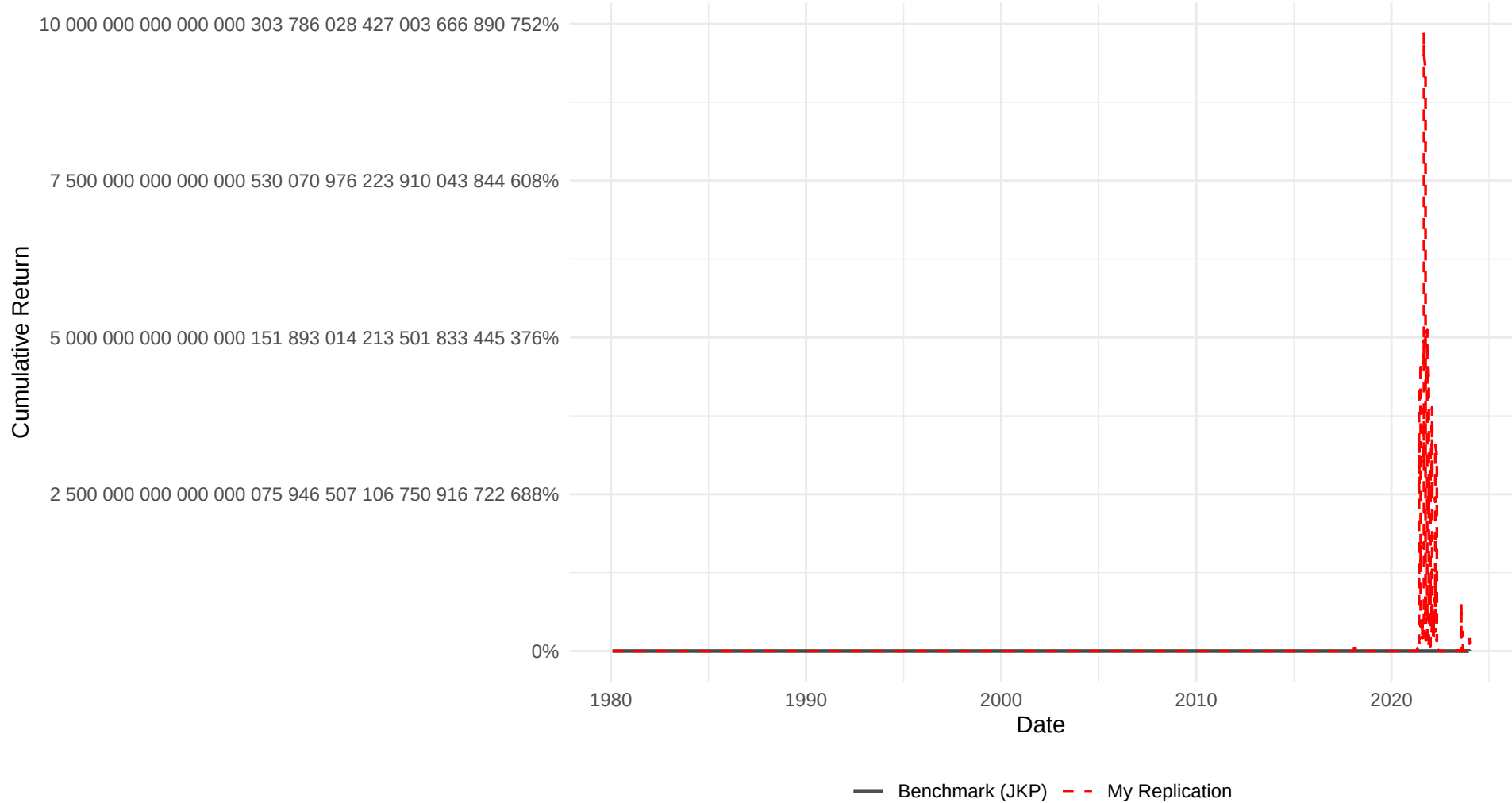
1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My Replica

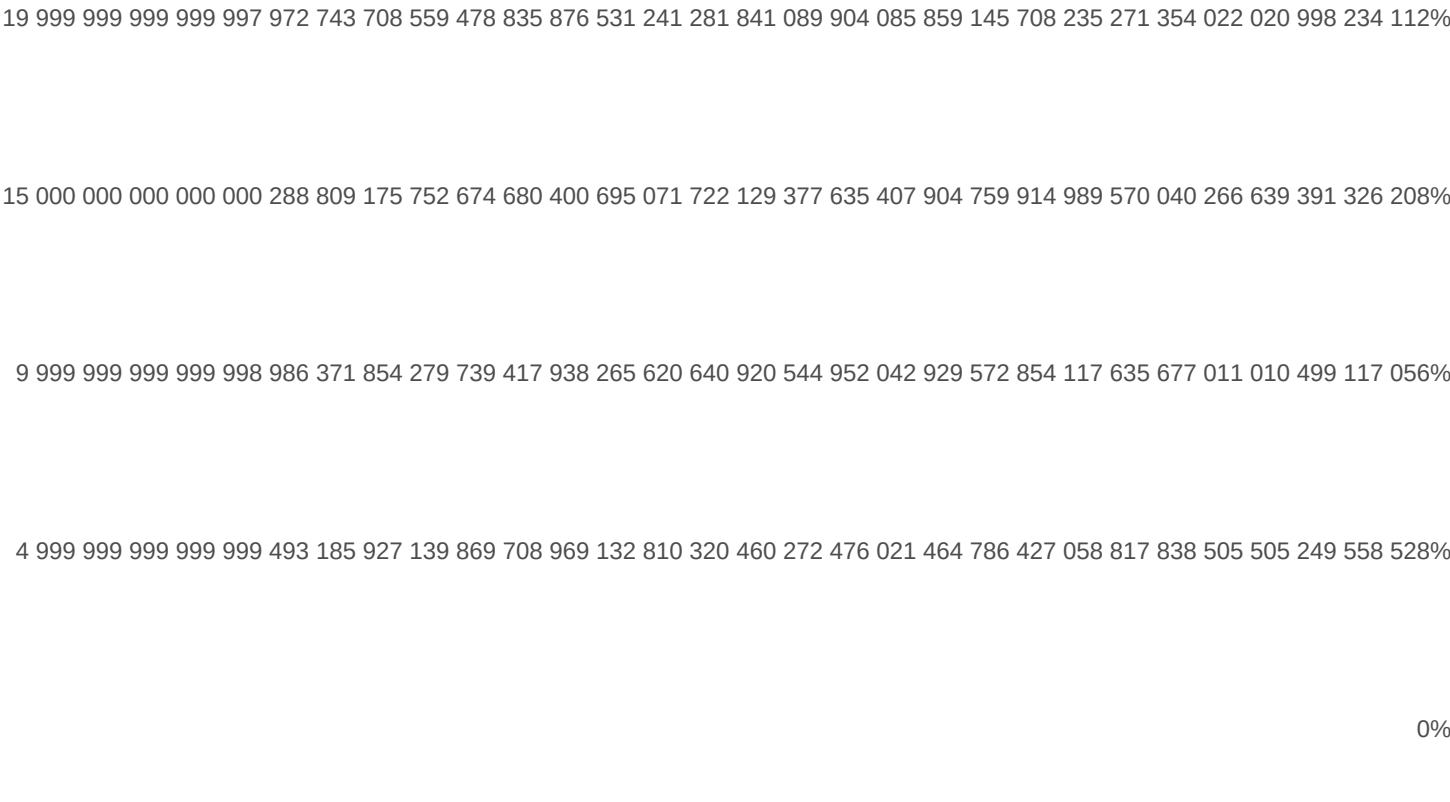
Monthly Corr: 0.1094 | Drift Check (Red should overlap Black)

Monthly Corr: 0.1094 | Drift Check (Red should overlap Black)



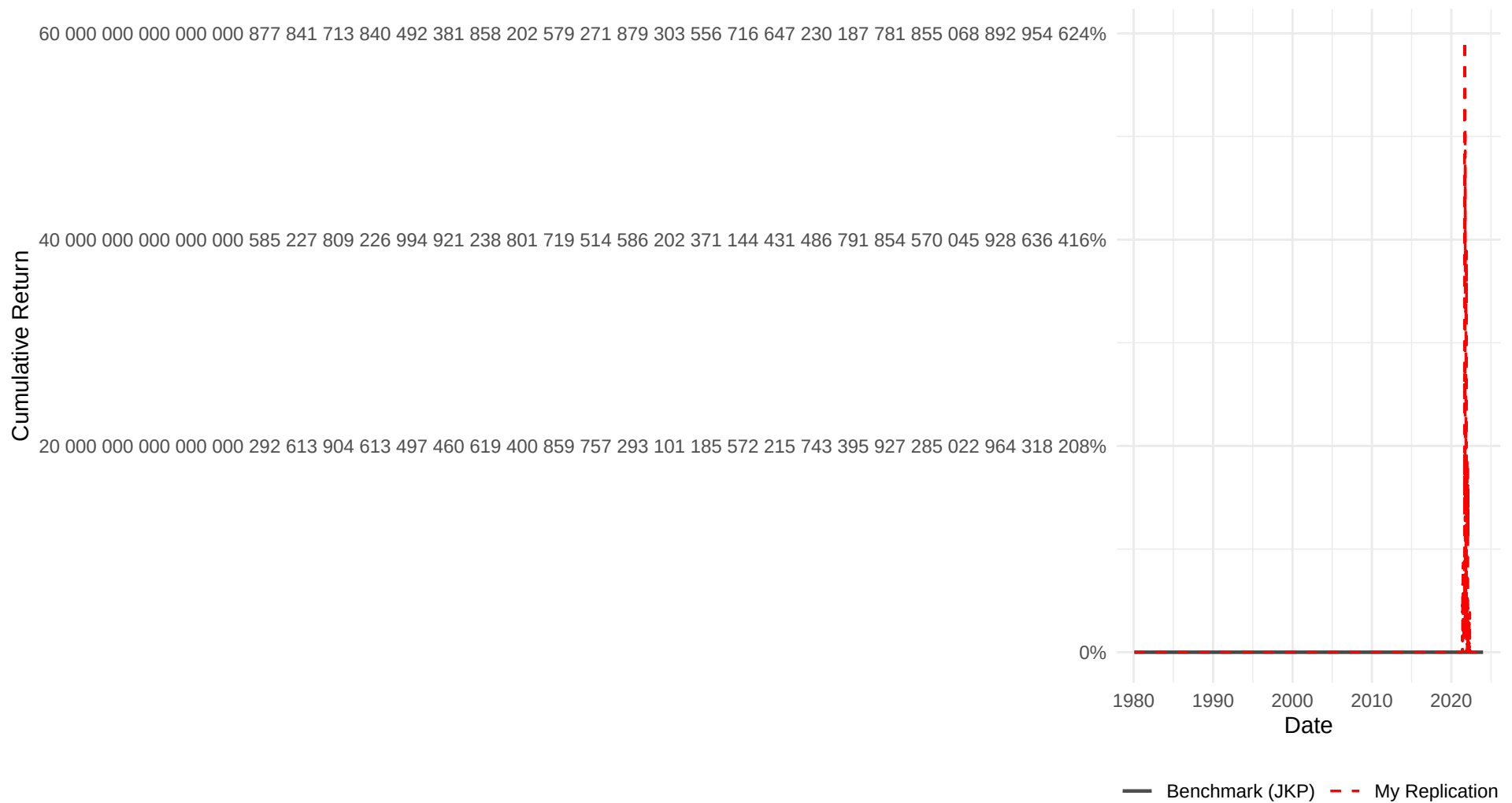
Factor: oaccruals_ni (C
Monthly Corr: 0.1234 | Drift Che

Cumulative Return



— Benchmark (JKP) - - My Replicati

Monthly Corr: 0.1287 | Drift Check (Re



Factor: ncol_gr1a (Cur

Monthly Corr: 0.1293 | Drift Ch

Cumulative Return

199 999 999 999 999 979 727 437 085 594 788 358 765 312 412 818 410 899 040 858 591 457 082 352 713 540 220 209 982 341 120%

149 999 999 999 999 977 558 572 236 863 829 055 100 797 746 570 813 933 451 269 901 990 276 512 069 056 164 662 916 153 344%

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicat

Factor: rd_sale (Cumulative Wealth)

Monthly Corr: 0.1309 | Drift Check (Red should overlap Black)

Cumulative Return

5 000 000 000 000 000 415 958 032 441 288 788 580 889 773 234 517 895 544 832%

4 000 000 000 000 000 193 386 768 462 214 636 230 113 579 383 562 057 023 488%

3 000 000 000 000 000 319 264 648 210 181 470 465 832 983 542 736 867 033 088%

2 000 000 000 000 000 096 693 384 231 107 318 115 056 789 691 781 028 511 744%

1 000 000 000 000 000 048 346 692 115 553 659 057 528 394 845 890 514 255 872%

0%

1980

1990

2000

2010

2020

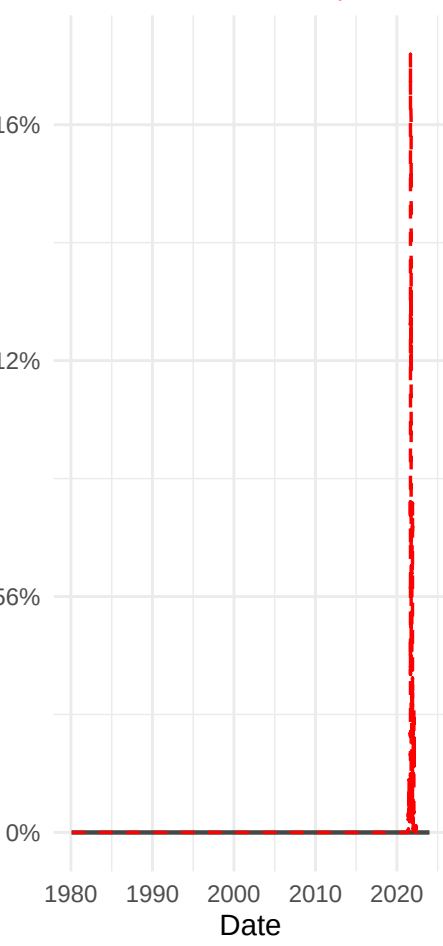
Date

— Benchmark (JKP) - - My Replication

Factor: taccruals_ni (C
Monthly Corr: 0.1424 | Drift Che

Cumulative Return

30 000 000 000 000 000 577 618 351 505 349 360 801 390 143 444 258 755 270 815 809 519 829 979 140 080 533 278 782 652 416%
19 999 999 999 999 997 972 743 708 559 478 835 876 531 241 281 841 089 904 085 859 145 708 235 271 354 022 020 998 234 112%
9 999 999 999 999 998 986 371 854 279 739 417 938 265 620 640 920 544 952 042 929 572 854 117 635 677 011 010 499 117 056%



— Benchmark (JKP) - - - My Replication

Factor: ret_

Monthly Corr: 0.

Cumulative Return

800 000 000 000 000 001 532 540 068 587 734 988 972 764 101 812 153 608 922 242 811 679 503 459 363 191 004 768 988 916 086 544 258 826 240%

599 999 999 999 999 937 492 030 790 988 111 045 840 645 313 566 047 673 833 294 806 699 119 762 143 004 211 252 326 069 460 636 125 888 512%

400 000 000 000 000 000 766 270 034 293 867 494 486 382 050 906 076 804 461 121 405 839 751 729 681 595 502 384 494 458 043 272 129 413 120%

200 000 000 000 000 000 383 135 017 146 933 747 243 191 025 453 038 402 230 560 702 919 875 864 840 797 751 192 247 229 021 636 064 706 560%

0%

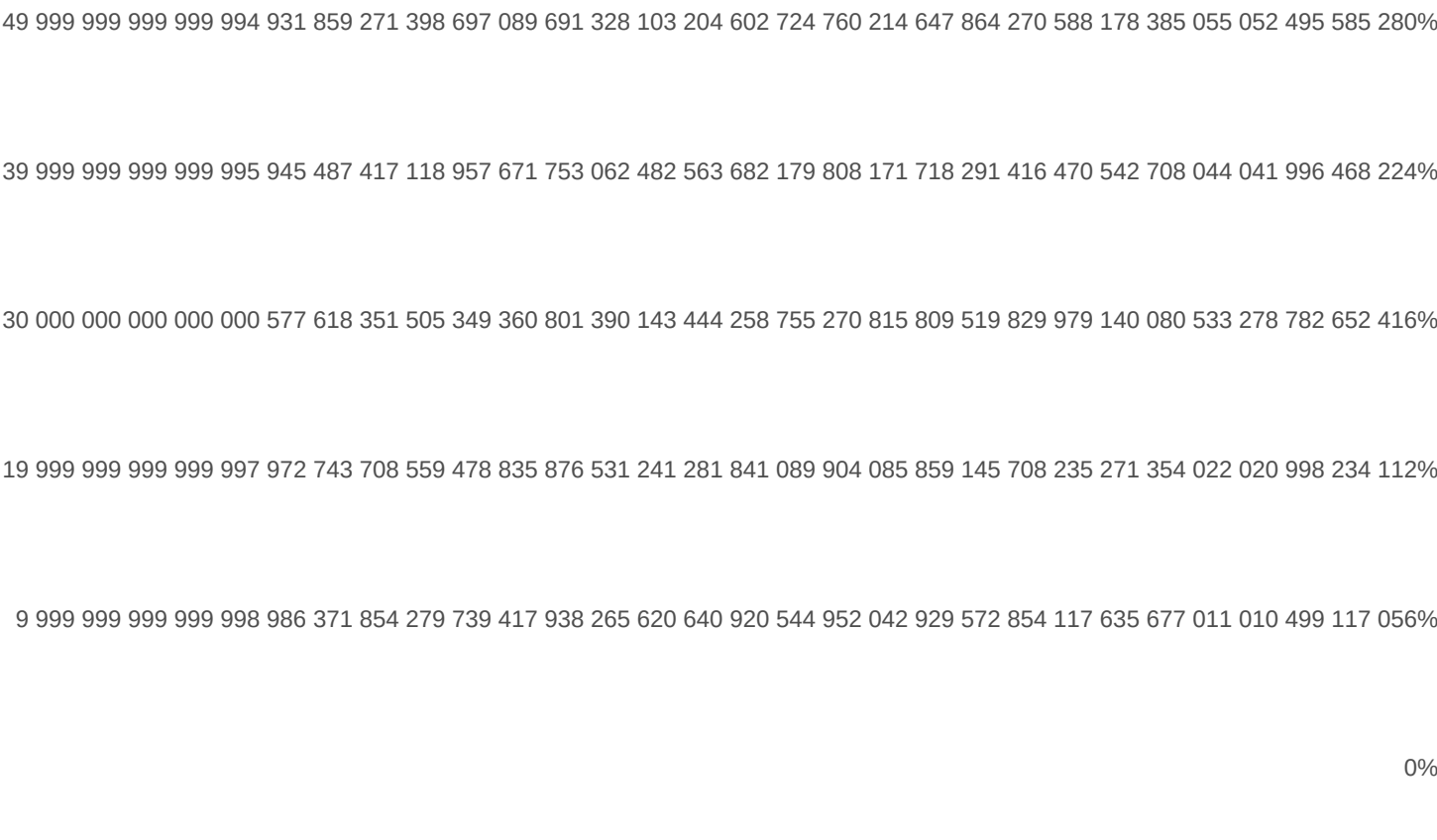
1980 1990 2000 2010 2020

Date

Benchmark (JKP) My

Factor: oaccruals_at (C
Monthly Corr: 0.1428 | Drift Che

Cumulative Return



— Benchmark (JKP) - - - My Replication

Factor: ami_126d (C

Monthly Corr: 0.1462 | Drift

Cumulative Return

50 000 000 000 000 001 010 943 956 357 797 349 428 804 816 160 717 887 056 888 842 810 400 200 249 908 215 467 934 891 376 640%

40 000 000 000 000 000 808 755 165 086 237 879 543 043 852 928 574 309 645 511 074 248 320 160 199 926 572 374 347 913 101 312%

30 000 000 000 000 000 606 566 373 814 678 409 657 282 889 696 430 732 234 133 305 686 240 120 149 944 929 280 760 934 825 984%

20 000 000 000 000 000 404 377 582 543 118 939 771 521 926 464 287 154 822 755 537 124 160 080 099 963 286 187 173 956 550 656%

10 000 000 000 000 000 202 188 791 271 559 469 885 760 963 232 143 577 411 377 768 562 080 040 049 981 643 093 586 978 275 328%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replica

Factor: cash_at (Cumulative)

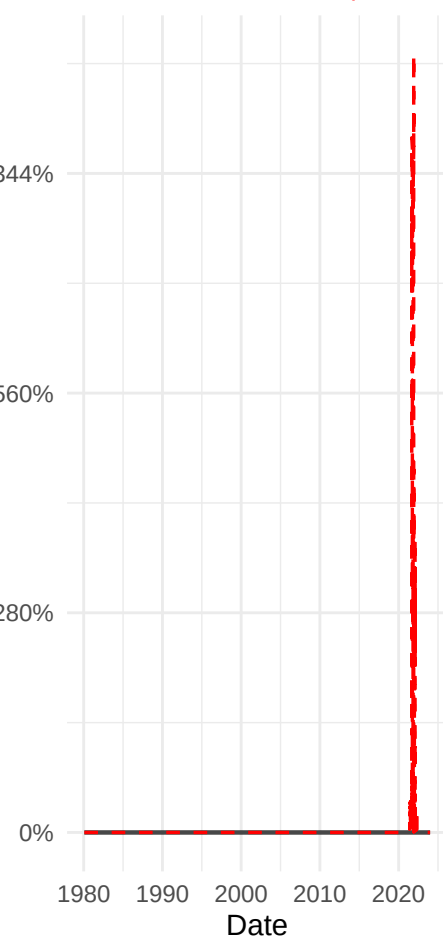
Monthly Corr: 0.1476 | Drift Ch

Cumulative Return

149 999 999 999 999 977 558 572 236 863 829 055 100 797 746 570 813 933 451 269 901 990 276 512 069 056 164 662 916 153 344%

99 999 999 999 999 989 863 718 542 797 394 179 382 656 206 409 205 449 520 429 295 728 541 176 356 770 110 104 991 170 560%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%



— Benchmark (JKP) - - My Replication

Factor: sale_emp_gr1 (Cumulative Wealth)

Monthly Corr: 0.1679 | Drift Check (Red should overlap Black)

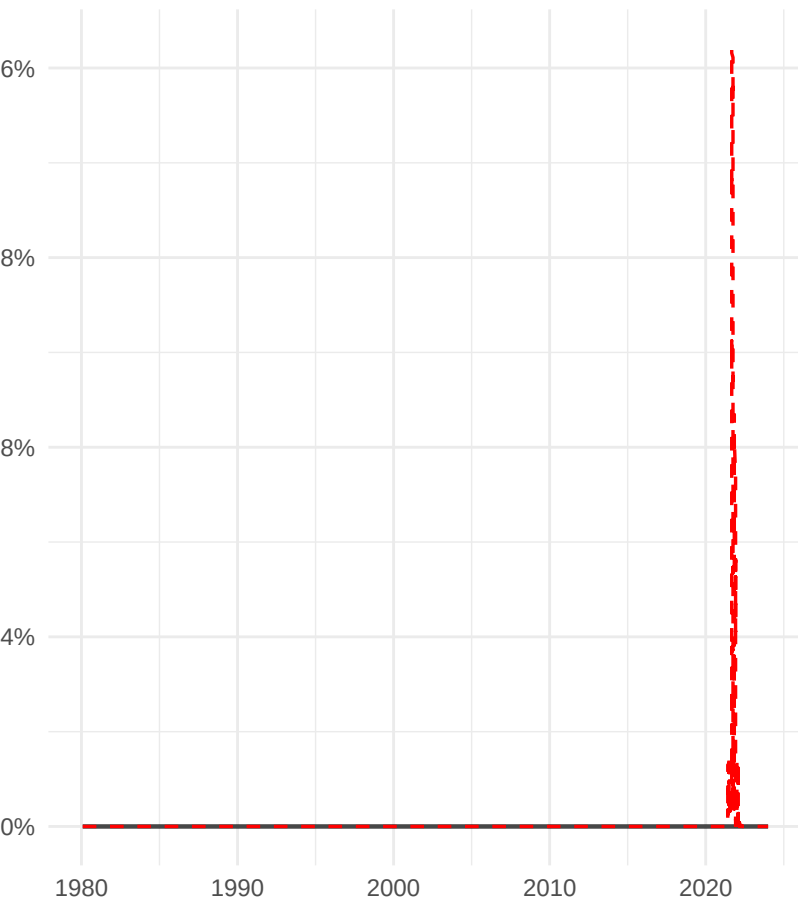
Cumulative Return

799 999 999 999 999 962 244 177 801 105 334 453 530 010 330 822 893 588 243 372 987 908 096%

599 999 999 999 999 947 737 890 524 799 487 428 298 335 448 893 589 197 139 730 956 812 288%

399 999 999 999 999 981 122 088 900 552 667 226 765 005 165 411 446 794 121 686 493 954 048%

199 999 999 999 999 990 561 044 450 276 333 613 382 502 582 705 723 397 060 843 246 977 024%



— Benchmark (JKP) - - - My Replication



Factor: taccruals_at (C
Monthly Corr: 0.1819 | Drift Che

Cumulative Return

74 999 999 999 999 988 779 286 118 431 914 527 550 398 873 285 406 966 725 634 950 995 138 256 034 528 082 331 458 076 672%

49 999 999 999 999 994 931 859 271 398 697 089 691 328 103 204 602 724 760 214 647 864 270 588 178 385 055 052 495 585 280%

24 999 999 999 999 997 465 929 635 699 348 544 845 664 051 602 301 362 380 107 323 932 135 294 089 192 527 526 247 792 640%

0%

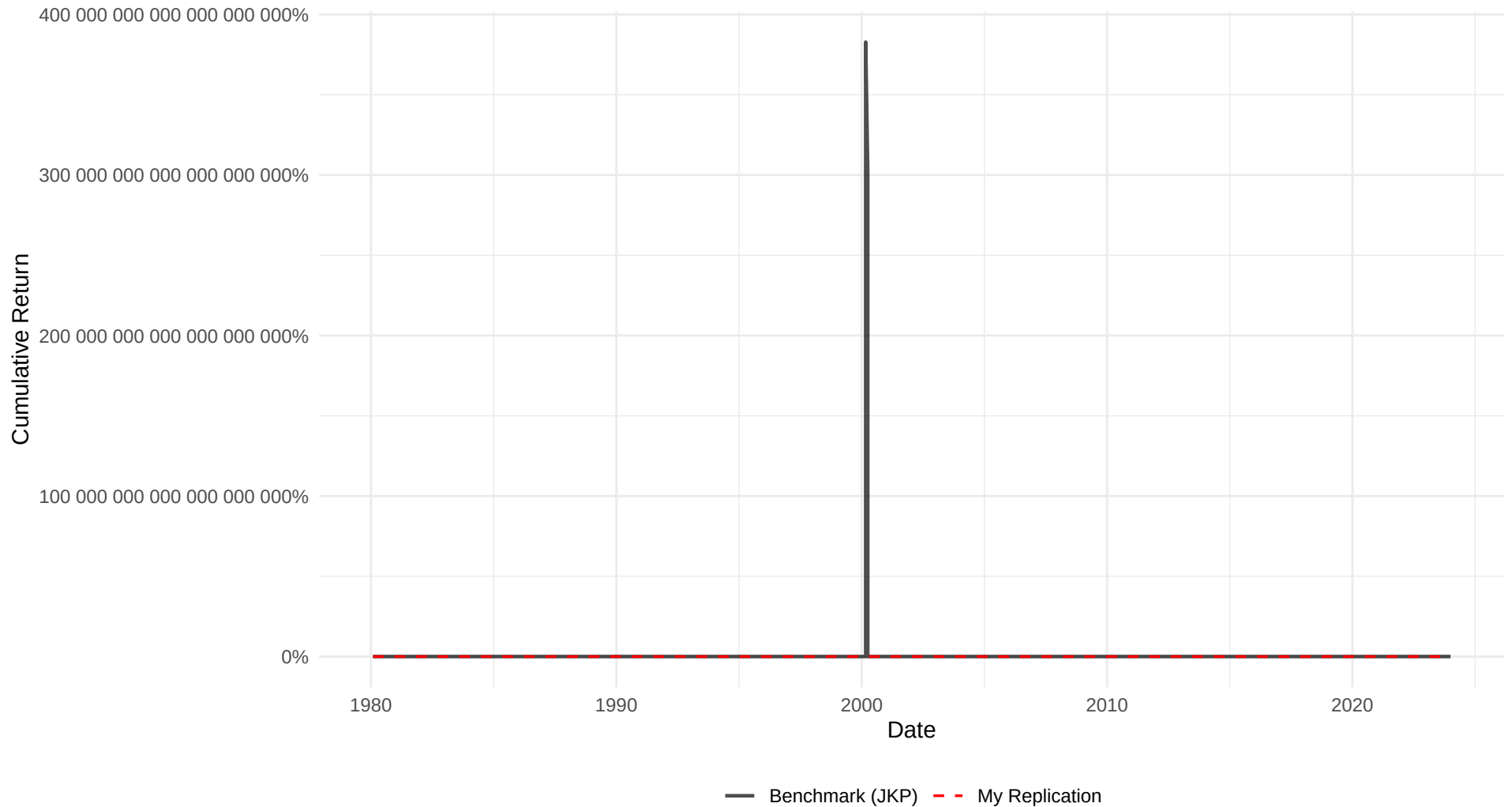
1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - My Replicati

Factor: age (Cumulative Wealth)

Monthly Corr: 0.1975 | Drift Check (Red should overlap Black)



Factor: tangibility (Cumulative)

Monthly Corr: 0.1993 | Drift Check

Cumulative Return

60 000 000 000 000 008 166 085 856 051 327 741 389 304 769 836 418 181 345 087 721 842 115 984 106 694 473 286 680 576%

39 999 999 999 999 998 376 668 978 254 014 509 259 679 843 585 938 057 586 789 560 392 268 156 918 038 177 020 641 280%

19 999 999 999 999 999 188 334 489 127 007 254 629 839 921 792 969 028 793 394 780 196 134 078 459 019 088 510 320 640%

0%

1980 1990 2000 2010 2020
Date

Benchmark (JKP) My Replication

Factor: rd_me (Cumulative Wealth)

Monthly Corr: 0.2191 | Drift Check (Red should overlap Black)

Cumulative Return

6 000 000 000 000 000 638 529 296 420 362 940 931 665 967 085 473 734 066 176%

4 000 000 000 000 000 193 386 768 462 214 636 230 113 579 383 562 057 023 488%

2 000 000 000 000 000 096 693 384 231 107 318 115 056 789 691 781 028 511 744%

0%

1980

1990

2000

2010

2020

Date

— Benchmark (JKP) - - My Replication



Factor: prc (C

Monthly Corr: 0.26

Cumulative Return

7 999 999 999 999 999 999 816 396 106 121 962 693 027 574 741 759 393 200 049 039 967 797 855 947 617 446 319 290 426 090 355 852 092 375 040%

6 000 000 000 000 000 000 608 281 934 206 151 983 003 754 428 539 776 160 187 464 202 044 413 536 873 780 704 807 558 813 285 689 132 580 864%

3 999 999 999 999 999 999 908 198 053 060 981 346 513 787 370 879 696 600 024 519 983 898 927 973 808 723 159 645 213 045 177 926 046 187 520%

1 999 999 999 999 999 999 954 099 026 530 490 673 256 893 685 439 848 300 012 259 991 949 463 986 904 361 579 822 606 522 588 963 023 093 760%

0%

1980 1990 2000 2010 2020

Date

— Benchmark (JKP) - - - My

Factor: ni_ivol (Cumulative

Monthly Corr: 0.2948 | Drift Check (R

Cumulative Return

200 000 000 000 000 002 926 139 046 134 974 606 194 008 597 572 931 011 855 722 157 433 959 272 850 229 643 182 080%

150 000 000 000 000 000 015 998 096 978 182 358 529 515 018 172 733 749 163 794 009 562 416 242 564 962 720 679 985 152%

100 000 000 000 000 000 001 463 069 523 067 487 303 097 004 298 786 465 505 927 861 078 716 979 636 425 114 821 591 040%

50 000 000 000 000 000 000 731 534 761 533 743 651 548 502 149 393 232 752 963 930 539 358 489 818 212 557 410 795 520%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication

Factor: bidaskhl_21d (Cur

Monthly Corr: 0.3115 | Drift Check

Cumulative Return

30 000 000 000 000 004 083 042 928 025 663 870 694 652 384 918 209 090 672 543 860 921 057 992 053 347 236 643 340 288%

19 999 999 999 999 999 188 334 489 127 007 254 629 839 921 792 969 028 793 394 780 196 134 078 459 019 088 510 320 640%

9 999 999 999 999 999 594 167 244 563 503 627 314 919 960 896 484 514 396 697 390 098 067 039 229 509 544 255 160 320%

0%

1980 1990 2000 2010 2020
Date

— Benchmark (JKP) - - My Replication